Predicting Urban Taxi Demand and Smart Charging Strategies for Electric Fleets

Advanced Analytics and Applications

Team 1

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Abstract

This project investigates how spatio-temporal demand analysis, machine-learning forecasts, and smart charging can support the operation of an electric taxi fleet in Chicago. The objective is to improve vehicle positioning and utilisation while ensuring that electric vehicles are charged reliably and at reasonable cost. Chicago taxi trips are used as a proxy for urban ride-hailing demand, defined as the number of observed pickups within a spatial unit and time window. The analysis combines taxi records, hourly weather observations, official spatial boundaries, and electricity-price data. After cleaning, the main dataset contains 14.47 million trips from January 2024 to May 2026. Trips are aggregated by community area or census tract and into hourly or four-hour intervals, allowing the consequences of different spatial and temporal resolutions to be examined.

The descriptive analysis reveals strong and recurring demand patterns. Taxi activity is lowest at 03:00 and peaks at 17:00, while weekdays, particularly Thursdays, are substantially busier than weekends. Spatial demand is even more concentrated: Near North Side, O'Hare, the Loop, and Near West Side together account for 71.06% of pickups with a reported community area. A Gaussian Mixture Model further identifies three operational hot spots: the extended city centre, O'Hare, and the Midway/Garfield Ridge area. Although these locations remain stable, their relative importance changes during the day. Central Chicago dominates daytime demand, whereas O'Hare becomes more important during evening and overnight periods. These findings suggest that fleet allocation should focus on a small set of high-volume zones and rebalance vehicles between the city centre and airports over the course of the day.

For demand prediction, Support Vector Regression and feedforward neural networks are trained at different spatial and temporal resolutions using calendar, weather, and spatial features. Their performance is compared with a transparent historical profile baseline that estimates demand separately for each spatial unit, time-of-day window, weekday, and month. This baseline already captures much of the stable demand structure and is difficult to outperform. Most pooled models achieve negative demand-weighted skill, indicating higher errors than the baseline in operationally important areas. The main pooled exception is the one-hour neural network, which reduces weighted RMSE by approximately 8% relative to the baseline. Dedicated nonlinear models also produce small improvements for selected high-demand areas. O'Hare remains particularly challenging because its continuous airport-related activity differs from the dominant citywide commuting pattern. Weather contributes comparatively little predictive information beyond the calendar profiles.

A separate reinforcement-learning experiment models home charging as a finite-horizon decision problem under uncertain energy demand. Deep Q-Network agents choose charging power in 15-minute intervals and are compared with a price-blind greedy policy. Using EPEX day-ahead prices, the price-aware agent reduces charging costs by 41.7% on episodes in which both strategies meet demand. However, its overall charging success rate is only 75.2%, compared with 98.9% for the greedy policy, showing that part of the saving results from insufficient reserve rather than better timing alone. The study therefore recommends a conservative operational architecture: use profile-based forecasts across most of the city, reserve specialised models for atypical high-demand zones, and introduce price-aware charging only with hard minimum-energy and departure-reliability constraints. The results constitute a proof of concept; deployment would require complete vehicle duty cycles, charging-infrastructure data, and more precise spatial observations.

Declaration on the Use of Generative AI

In accordance with the course's requirements on the disclosure of generative AI use, we declare that generative AI tools were used during the preparation of this project, in the following areas:

- **Coding**: Assisting with writing, debugging, and refactoring code in the analysis notebooks (data collection, descriptive analytics, predictive analytics, and reinforcement learning) and supporting scripts.
- **Writing**: Drafting sections of this report and supporting documentation, including explanatory text and descriptions of methods and results.
- **Rewriting/editing**: Revising, restructuring, and improving the clarity, grammar, and conciseness of previously written text and code produced by the team.

All analytical decisions, model design choices, interpretation of results, and conclusions presented in this report reflect the judgment of the team members. Generative AI outputs were reviewed, verified, and edited by the team before inclusion; the team takes full responsibility for the accuracy and integrity of the final content.

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1 Introduction

Urban taxi demand is highly uneven across both space and time. In a city such as Chicago, trip volumes vary by neighborhood, hour of day, weekday, weather conditions, airport activity, and commuting patterns. For a fleet operator, this creates a practical planning problem: vehicles should be available where passengers are likely to request rides, while unnecessary idle time, empty repositioning, and inefficient charging should be reduced.

This project uses Chicago taxi trips as a proxy for urban ride-hailing demand. The core problem is to understand and predict how many trips will start in different parts of the city during a given time window. In this report, demand is therefore defined as the number of taxi pickups per spatial unit and time interval.

1.1 Business Goal

The business goal is to support better operational decisions for a taxi or ride-hailing fleet operator. More accurate knowledge of where and when demand occurs can help operators position vehicles in high-demand areas, reduce passenger waiting times, avoid unnecessary idle time, and improve the utilization of available vehicles. For an electric fleet, this planning task is connected to charging decisions: vehicles need enough energy to serve expected demand, but charging should be scheduled in a cost-efficient way.

From a managerial perspective, the project aims to turn historical trip data into practical recommendations for fleet positioning, demand planning, and charging strategy. The final value of the analysis is not only whether a model predicts demand accurately, but whether the results can inform decisions that improve service reliability and operational efficiency.

1.2 Data Mining Goal

The data mining goal is to build a spatio-temporal demand prediction task from raw taxi trip records and external contextual data. Individual trips are aggregated into panels by time window and spatial unit, and the target variable is the number of pickups in each panel cell. Calendar variables, weather data, and spatial identifiers are used as explanatory features.

The project evaluates whether machine learning models can predict aggregate taxi demand at useful resolutions. In particular, Support Vector Machine models and neural networks are trained and compared against simple time-profile baselines. Model performance is assessed using appropriate error metrics and benchmarking, with attention to whether the models perform well not only overall but also in the high-demand areas that matter most for fleet operations. In a second analytical task, a reinforcement learning approach is used to study how charging decisions for an electric taxi can be optimized under operational constraints.

2 Data Description

The analysis combines taxi trip records, hourly weather observations, spatial reference data, and electricity prices. Taxi and weather data cover 2024-01-01 through 2026-05-31; the period ends in May because the city portal reports recent trips with a delay, so including the incomplete June data would create an artificial decline in demand.

Taxi trip data. The main source is the Chicago Taxi Trips dataset from the City of Chicago Data Portal (City of Chicago, 2026e) (dataset `Taxi Trips (2024-)`). Each row is one reported trip by a licensed Chicago taxi; before cleaning the extract contains 16,086,196 rows and 21 columns, with trip timestamps and duration, distance, fare components, payment type, taxi company, anonymized identifiers, and pickup/dropoff locations. Locations are reported through the identifiers and centroids of Chicago's 77 community areas and 878 census tracts. Census tracts provide finer spatial detail but are frequently masked in low-volume areas for privacy; where a tract is unavailable, the coordinate columns may instead hold the community-area centroid. This mixed spatial resolution limits fine-grained interpretation and is handled during data preparation and spatial analysis.

Weather data. Hourly weather observations come from the Open-Meteo archive API (Open-Meteo, 2026): 21,168 observations and nine variables over the same period as the taxi data, including temperature, apparent temperature, precipitation, snowfall, wind speed, cloud cover, and WMO weather conditions. Weather is joined to the demand data by timestamp and provides exogenous features that are also known for future prediction periods.

Spatial reference data. Official boundary files provide polygons for Chicago's 77 community areas (City of Chicago, 2026c), 878 census tracts (City of Chicago, 2026a), and the city boundary (City of Chicago, 2026b), supporting mapping, spatial joins, and demand aggregation at different resolutions. Selected spatial visualizations use OpenStreetMap basemap tiles (OpenStreetMap contributors, 2026) for visual context; map data are © OpenStreetMap contributors, available under the Open Database License.

Electricity price data. The reinforcement-learning analysis uses EPEX SPOT day-ahead prices from the Fraunhofer ISE Energy-Charts API (Fraunhofer Institute for Solar Energy Systems ISE, 2026), covering the German–Luxembourg bidding zone from January through March 2025 and converted from EUR/MWh to EUR/kWh. For each of the 90 days, prices between 14:00 and 16:00 are transformed into eight 15-minute values matching the charging decisions; these published prices are observable to the price-aware charging agent, while the price ultimately billed is simulated by adding a random short-term deviation (details in the reinforcement-learning section).

Granularity and limitations. The cleaned taxi data contain 14,465,662 rows and 27 columns. For spatial analytics and predictions, individual trips are aggregated into spatio-temporal demand panels by community area or census tract and hourly or four-hour window. Key limitations are privacy-related location masking, the reporting lag in recent portal data, mixed spatial precision, and the use of licensed taxi trips as a proxy for wider ride-hailing demand.

3 Data Preparation

The raw taxi export contains 16,086,196 rows across 21 columns. The cleaning pipeline concatenates the 2024, 2025, and 2026 CSV exports, applies the steps below, and writes the result as a single Parquet file. In total, 10.1% of trips (1,620,534 rows) are removed.

Duplicate removal. Trips are deduplicated on `trip_id`, keeping the first occurrence (775 rows); ID-less rows are set aside first and rejoined afterward, so `drop_duplicates` does not collapse several genuine but ID-less trips into one.

Missing spatial identifiers. Location is reported at two resolutions: `census_tract`, masked for over half of trips because the city hides low-volume tracts for privacy, and `community_area` (98.2% pickup, 91.6% dropoff complete, missing only outside city limits). Out-of-city trips are kept, with `pickup_flag`/`dropoff_flag` columns marking area presence so later spatial analyses can filter without dropping rows.

Resolution-consistent centroids. The raw centroid columns mix resolutions – tract centroid where available, else community-area centroid. A clean area-to-centroid mapping, derived from the rows already at area-centroid resolution, is joined back as separate `*_ca_centroid_lat/lon` columns for consistent area-level aggregation.

Incomplete records. Trips missing `fare`, `trip_total`, `trip_seconds`, or `trip_miles` are dropped (34,687 rows), since no meaningful analysis is possible without them.

Outlier and plausibility filtering. Trips below Chicago's $3.25 minimum fare (City of Chicago, 2026d) or shorter than 60 seconds, and those above the 99.9th percentile of distance (43.0 miles) or duration (165 minutes), are removed – 1,585,072 rows, the bulk of all removals. Negative `tips/tolls/extras` would also be dropped, but none occur; zero is legitimate (e.g. cash trips that report no tip).

`payment_type` and `company` are left untouched; their missing values remain `NA` and are handled per analysis. The cleaned dataset has 14,465,662 rows and 27 columns, spanning 2024-01-01 to 2026-05-31. Table 1 in the appendix gives the full column-by-column reference: dtype, completeness, distinct-value count, and description for every column.

4 Descriptive (Spatial) Analytics

This section analyzes 14,465,662 cleaned taxi trips from January 2024 to May 2026. Demand is defined as observed pickups and therefore represents served taxi demand rather than all latent ride-hailing demand.

4.1 Temporal and Trip Patterns

Demand follows a clear daily and weekly cycle. The lowest-volume hour is 03:00, with an average of 62.96 pickups per observed day, while demand peaks at 17:00 with 1,194.63 (Figure 1). Thursday is the busiest day (18,856 trips per day) and Sunday the quietest (12,591; Table 2). Weekend demand is lower overall and relatively stronger around midnight, but neither a strict Saturday–Sunday nor an extended Friday–Sunday definition produces a separate late-night peak; both still reach their maximum at 17:00. Commuting, business activity, and after-work travel are plausible explanations for the weekday and late-afternoon concentration.

Temporal aggregation changes the operational signal (Figure 2). Hourly bins retain the exact 17:00 peak and are useful for short-term vehicle positioning, whereas wider bins smooth local fluctuations. Four-hour windows retain the broader 16:00–20:00 demand peak and provide a more stable signal for fleet allocation.

Trip distance, duration, and price are strongly right-skewed (Figure 3; Table 3). A typical trip has a median distance of 3.85 miles, lasts 16 minutes, and has a metered fare of USD 15.25. The corresponding means are higher because a smaller number of long and expensive trips pull them upward. Trip composition also varies by hour: median distance and mean fare are lowest at 08:00, whereas median distance is highest at 05:00 and mean fare peaks at 04:00 (Figure 4). Short urban commuting trips may explain the morning trough, while a larger share of airport and cross-city trips is a plausible explanation for the early-morning peaks. These interpretations are descriptive and do not isolate location, surcharges, or traffic effects.

Idle time is not directly recorded in the dataset. We therefore approximate it as the time between the end of a taxi's recorded trip and the start of its next recorded trip. Negative gaps are excluded as implausible observations. Under the primary four-hour cutoff, gaps exceeding four hours are excluded because they are more likely to represent off-shift periods or incomplete trip records than regular between-trip idle time. This rule retains 12,072,245 transitions, corresponding to 84.76% of all non-negative consecutive-trip gaps. The retained gaps have a mean of 44.49 minutes and a median of 30 minutes. Since the resulting estimate depends on the selected cutoff, alternative thresholds are reported in Table 4 and visualized in Figure 5. The measure should therefore be interpreted as a proxy for between-trip inactivity rather than a direct measurement of vehicle availability or utilization.

4.2 Spatial Patterns and Resolution

Pickup and drop-off demand are concentrated in central Chicago and at the airports (Table 5). Near North Side leads both pickups (3,264,625) and drop-offs (3,528,594). O'Hare is strongly origin-oriented, with 2,985,521 pickups but only 579,319 drop-offs. Together, Near North Side, O'Hare, the Loop, and Near West Side account for 71.06% of pickups with a reported Community Area. Downtown employment, hotels, tourism, nightlife, rail connections, and airport flight schedules are plausible demand generators.

The leading zones vary over time (Table 6). Near North Side dominates pickups from 04:00–20:00, while O'Hare leads from 00:00–04:00 and 20:00–24:00. Near North Side leads most drop-off blocks, with the Loop leading from 04:00–08:00. Origins and destinations must therefore be modeled separately, and fleet positioning should change over the day.

Most trips have both their pickup and drop-off assigned to a Chicago Community Area across every weekday (Figure 6). The smaller comparison group contains trips for which at least one endpoint is outside Chicago or cannot be assigned unambiguously, so it should not be interpreted as a direct count of cross-boundary travel.

Three different spatial resolutions were explored. Community areas are nearly complete, recognizable operating zones but hide local variation (Figure 7). Census tracts add finer spatial detail: pickup tract identifiers are available for 44.6% of trips, while the descriptive tract maps use the 6,245,422 trips (43.17%) for which both pickup and drop-off tract identifiers are available, since low-volume tracts are masked for privacy (Figure 8). H3 resolution 7 offers a uniform grid (Figure 9), but the anonymized data carry only 573 distinct pickup and 687 drop-off coordinate pairs, so mapping them to hexagons would invent spatial detail the source does not contain. Because demand is natively reported at community-area and census-tract resolution, H3 is kept only as a descriptive view and dropped for prediction, where a mismatched grid would distort the picture.

4.3 Gaussian-Mixture Hot Spots

A Gaussian Mixture Model (GMM) complements the discrete maps with a parametric spatial density estimate. It is fitted to 2,787,934 tract-resolved pickups from 2025 at 421 locations inside Chicago. Each component provides a center, spatial extent, and share of pickup demand. A 100-meter covariance floor prevents components from collapsing onto anonymized centroids.

The BIC curve has an elbow at six components, but the final model uses three operationally interpretable and spatially separated zones: the extended city center (76.3% of tract-resolved pickups), O'Hare (21.0%), and Midway/Garfield Ridge (2.6%). Additional components mainly divide the contiguous downtown area near the approximately 600-meter resolution limit. The fitted density and component estimates are provided in Figure 10 and Table 7.

The hot-spot locations remain stable over the day, but their importance changes (Figure 11). The city-center share is approximately 80% during 07:00–19:00 and 61% at night; O'Hare rises from 18% to 36%. This supports rebalancing between a stable set of downtown and airport zones rather than uniform city-wide coverage.

5 Predictive Analytics

We predict taxi pickup demand – the number of trips originating in a spatial unit within a fixed time window – with two model families: Support Vector Regression (SVR) and feedforward neural networks (NN). Both are evaluated at community-area resolution in 4-hour and 1-hour windows; census tract (4-hour) is run with the neural network only, all due to computational cost. Community area at 4-hour resolution is the default throughout, and the other resolutions are presented as deviations from it.

5.1 Modeling Approach

Both families share one task framing: a single **pooled model per experiment**, trained across all spatial units at once, with spatial identity encoded as one-hot dummies rather than as separate per-unit models. The families are complementary: linear SVR is interpretable through its coefficients, kernelized SVR captures nonlinear relationships (e.g. between spatial identity and time), and feedforward networks learn them far more flexibly, but at the cost of interpretability.

The central obstacle is that demand is not one distribution: it varies roughly three orders of magnitude across community areas, and each unit has its own daily and weekly rhythm – an airport runs around the clock on flight schedules, a downtown office district peaks at commute hours and goes quiet overnight – yet a pooled model must represent all of it with one parameter set. This is hardest for a linear SVR, whose single global coefficient vector can find only one compromise direction across every unit; a kernel SVR could represent the heterogeneity in principle, but its quadratic-to-cubic training cost forces the kernel variants onto a 10,000-row subsample. A neural network can absorb the heterogeneity by learning unit-specific nonlinear interactions, at the cost of complexity and interpretability.

Both use the same exogenous features: weather (temperature, precipitation, snowfall, wind, cloud cover), cyclical time encodings (sine/cosine pairs for hour-of-day and weekday, two harmonics each), month, and weekend/holiday indicators. No lagged or autoregressive demand features are used, since the goal is to predict from spatial identity and time rather than recent demand. Both regress on `log1p(count)` and back-transform via a clipped `expm1`, as raw demand is heavily right-skewed (Figure 12).

5.2 Validation Strategy

Every model is judged against a **profile baseline**: the train-set mean demand for each (time-of-day window, weekday, month) bucket, computed separately per spatial unit (Equation 1). It is a natural reference because it is itself deployable – a cheap, interpretable lookup of each unit's historical averages – so a model that cannot beat it is not worth the added complexity. Being fitted per unit, it sidesteps the heterogeneity problem above, never compromising one parameter set across areas whose demand differs by three orders of magnitude, which makes it a demanding reference throughout. Both families use the same chronological holdout: training on 2024-01-01 through 2025-12-31 and testing on 2026-01-01 through 2026-05-31, over a complete zero-filled grid of every unit and time window, so evaluation includes genuine zero-demand periods, not only observed trips. Metrics are computed over the whole test period; for the qualitative trace figures, a single representative week – beginning 2026-03-02, referred to throughout as the **March evaluation week** – shows each model's predictions against actual and baseline demand at readable resolution.

The baseline already explains much of the variation in the data. At the default community-area, 4-hour resolution its pooled test R^2 is 0.933 (Table 8), tracking the daily and weekly cycle closely across the March evaluation week (Figure 13); the same holds at finer settings (pooled R^2 0.918 at 1-hour and 0.896 at census-tract level). Per-unit fit is far more variable – median per-area R^2 is only 0.208 at 4-hour resolution, with 21 of 77 areas negative (Figure 14) – because the pooled figure is dominated by the few highest-volume units. Most of that spread is irreducible, though: in low-demand units (around one trip per window) demand is largely count noise, so per-area R^2 is low for any predictor.

Two headline metrics summarise each model, both built to survive the demand heterogeneity above. **Demand-weighted skill** measures how far a model beats the profile baseline, one unit at a time: for a unit i it is one minus the ratio of the model's mean absolute error to the baseline's MAE on that same unit (Equation 2), so 0 ties the baseline, 1 is zero error, negative is worse and positive is better. The per-unit skills are then collapsed into one number weighted by each unit's median demand (Equation 3), so the busiest areas dominate – deliberately, since an unweighted mean would let the dozens of near-empty units outvote the high-volume areas that drive fleet allocation. **Weighted RMSE** (Equation 4) is the absolute-scale companion: a single pooled root-mean-square error over every test row, in trips per window so it reads directly rather than as a ratio, with each squared error weighted by its area's train-period mean demand, so a miss in a high-volume area counts far more than the same miss in a low-demand one. Being an absolute magnitude rather than a comparison against the baseline, it is the metric used to rank the two model families head-to-head.

5.3 Results

5.3.1 Support Vector Regression – Community Area, 4-Hour

Nine SVR configurations were run at this resolution, forming a progression that steadily relaxes the single pooled-model constraint; all are collected in Table 9, with the grids used to tune the kernel models in Table 11. The plain pooled `LinearSVR` reaches demand-weighted skill −0.827 (weighted RMSE 254.94), far below the baseline, because one global coefficient vector cannot serve areas whose demand spans three orders of magnitude. Two attempts to correct this while keeping the pooled linear model both fail: weighting each training row by its area's demand (the *demand-weighted* variant, −0.675), and splitting the 77 areas into three demand terciles each with its own `LinearSVR` (the *demand-tiered* variant, −0.732), leave skill essentially level with the unmodified fit.

Allowing nonlinearity helps but is not enough: an RBF and a polynomial kernel, each tuned by grid search on a 10,000-row subsample (the full panel is too costly for kernel SVR), reach −0.232 and −0.221 – a clear step up from the linear variants, yet still short of the baseline. Only abandoning pooling closes the gap: fitting separate models for the 20 areas above a median-demand threshold of 10 trips per window, the tuned per-area RBF SVR is the single configuration with positive demand-weighted skill, +0.012 (weighted RMSE 143.80) – the only SVR to beat the baseline, and only once pooling is dropped. Figure 15 shows the pooled variants over the March week in Community Area 8, and Figure 16 maps pooled `LinearSVR` skill across the 77 areas.

5.3.2 Neural Network – Community Area, 4-Hour

Two networks were run: an untuned shallow network (one hidden layer) and a deeper network selected by a one-factor-at-a-time architecture search over the hyperparameters in Table 12; test scores are in Table 10. The shallow network reaches demand-weighted skill −0.028 (weighted RMSE 144.96) and the tuned deep network −0.027 (145.93), against the baseline's 139.02 – both just below the baseline. Figure 18 traces the three models across the March evaluation week in the highest-demand area (Community Area 8), and the per-area skill map (Figure 19) is at or above zero across most of the city, with O'Hare (community area 76) the one clearly negative area (skill −0.227). Adding O'Hare-specific time features (the area dummy interacted with the hour-of-day harmonics) was tried but did not help – the best variant reached only −0.030 (Table 13, Figure 20).

5.3.3 Spatial and Temporal Resolution Variations

Community area, 1-hour. The same 77 areas are now split across 24 hourly buckets instead of 6, quadrupling the rows and the baseline's granularity; scores are in Table 14 (SVR) and Table 15 (NN), each against the profile baseline. For SVR, the pooled `LinearSVR` falls to −1.016 and the best tuned kernel (RBF) reaches −0.331. For the NN, the tuned deep configuration from 4-hour (reused unchanged, for cost) reaches +0.062 (weighted RMSE 36.26 vs the baseline's 39.46) – the only positive NN result in the study – while the shallow network stays negative (−0.048). Traces and per-area skill maps are in Figure 21, Figure 23, Figure 22 and Figure 24.

Census tract, 4-hour. Only 44.6% of trips carry a true tract-level location; Chicago masks the rest to the community-area centroid when a tract has too few trips, for privacy. To recover realistic tract-level demand, tract-precision coverage was measured per community area, and only areas retaining at least 5% of their trips at tract precision were kept – a natural gap separates areas at 4.85% from those at 7.17% – leaving **179 of 878 tracts across 11 community areas** in scope (Figure 25); the rest are left out of tract-level modelling. Within scope, per-window counts were built from tract-precision trips only and scaled up by each area's inverse retention rate, extrapolating the observed fraction to each tract's total demand; this keeps masked points from spiking demand onto whichever tract holds a community area's centroid.

Only neural networks were run here (Table 16): the shallow network reaches −0.098 (weighted RMSE 113.77 vs the baseline's 100.99) and the deep network −0.340; traces and the per-tract skill map are in Figure 27 and Figure 28. SVR was not run at tract level because a per-tract one-hot encoding (up to 878 dummy variables) is computationally infeasible for kernel SVR.

5.4 Discussion

5.4.1 Model Comparison

Across every setting the profile baseline is a strong reference, precisely because it is fitted per unit: a single pooled model must instead reconcile demand spanning three orders of magnitude and areas with distinct daily and weekly rhythms, a compromise the baseline never faces. This difficulty is not distributed evenly across the high-demand areas. The pooled models predict the busy, commute-driven areas such as downtown well – at 4-hour resolution the deep network matches or beats the baseline (Figure 19), and at 1-hour resolution it is clearly positive there (Figure 24). The exception is O'Hare; it follows a different daily pattern from the citywide commute cycle and is the single worst area for the deep network (skill −0.227 at 4-hour resolution) and – as one of the highest-demand units – on its own pulls the network to near break-even overall. The neural network comes closest to the baseline, reaching demand-weighted skill +0.062 at 1-hour resolution; pooled SVR trails at every resolution, and only abandoning pooling – a dedicated per-area RBF model for the 20 busiest areas – turns positive (+0.012).

A second source of error is shared by every model and visible in Figure 29: demand is not stationary over the observation window. Across the January–May span common to all three years, average daily demand rose +9.0% from 2024 to 2025 and a further +4.7% into 2026, every one of the five months higher year-on-year. Because the models train on 2024–2025 and test on 2026, and the features carry no year or trend term (nor any autoregressive signal), every predictor – the baseline included – is calibrated to the lower 2024–2025 level and under-predicts 2026. Part of the actual-minus-predicted residual is thus this unmodelled upward trend, not model misfit, on top of the per-area effects above.

One input the models have but the baseline lacks is weather, and it is a weak signal. In the pooled `LinearSVR` the calendar features dominate – hour-of-day above all – while every weather variable ranks near the bottom, temperature strongest, precipitation and snowfall negligible (Figure 17). This is expected: weather is citywide, so it can only shift the whole city up or down, not separate one area from another, and the baseline's month bucket already absorbs the seasonal part.

Resolution matters in opposite directions: for matched SVR types skill is worse at 1-hour than 4-hour (the finer grid sharpens the high-demand gap), while for the network the 4-hour architecture transfers to 1-hour cleanly and produces its one clear win. Census-tract results are weakest overall: even with inverse-retention extrapolation the per-unit signal is thin once 179 units compete for demand, and both networks fall below the tract baseline.

5.4.2 Practical Implications

The overriding practical finding is that the profile baseline is hard to beat. Across the low-demand tail – most of the city's units – absolute errors are only a fraction of a trip per window, so no model's edge over the cheap baseline is operationally meaningful. What value a trained model adds is confined to the atypical high-demand hubs.

Whether that value justifies training at all is a close call. At the default 4-hour resolution no pooled model beats the baseline; the only configuration that does is the per-area RBF, and only by +0.012. The neural network is the strongest of the trained models – pooled, it beats the pooled RBF at both resolutions (−0.027 vs −0.232 at 4-hour) and is the only pooled model ever to beat the baseline – yet its own advantage shows up only at 1-hour resolution, where weighted RMSE falls just from 39.46 to 36.26 (about 8%). The recommendation is therefore to run the cheap baseline, or a single pooled network, across the city, and add a per-area or higher-resolution model only for the few high-demand hubs whose pattern departs from the citywide norm.

6 Reinforcement Learning

The task considers a taxi driver who charges her vehicle at home during a fixed afternoon window; a flat charging rate is replaced by an agent that selects the charging power for each interval, subject to the vehicle's battery capacity. The battery must cover a shift's demand, which is stochastic and unknown until departure, so recharging cost must be weighed against a heavy penalty for arriving under-charged. The resulting environment assumptions are summarized in Table 17.

6.1 Markov Decision Process

The charging problem is modeled as a finite-horizon Markov Decision Process. Without day-ahead price information, the state is $s_t = [SOC_t, t, C]$, where SOC_t is the current energy level, $t \in \{0, \dots, 8\}$ the time step, and C the battery capacity; the realized demand is deliberately excluded. When day-ahead prices are included, the state extends to $s_t^{EPEX} = [SOC_t, t, C, \pi_t^{DA}, \bar{\pi}_{\geq t}^{DA}]$, adding the current and mean-remaining day-ahead price.

Four discrete charging powers of 0, 5, 11, and 22 kW are used initially; because this sparse spacing leads to degenerate behavior under the exponential cost function, a refined grid of 0, 4, 8, 12, 16, 20, and 22 kW is used in the final iteration. For a 15-minute interval the transition is $SOC_{t+1} = \min(C, SOC_t + p_t \cdot 0.25)$. The step reward is the negative charging cost; after the eighth decision a terminal penalty is added if $SOC_8 < d$, giving $r_8 = -\mathrm{cost}_8 - 100 \max(0, d - SOC_8)$. A cost-reliability trade-off results: additional energy is inexpensive relative to a shortage, while the tail risk must be learned from sampled experience.

6.2 Cost Environments and Electricity Prices

Two `gymnasium` environments are used, sharing the same battery dynamics and hidden-demand process but differing in charging-cost model. In the stylized environment, the cost of selecting power p in interval t is $\mathrm{cost}(t, p) = \alpha_t \exp(p)$, where α_t increases from 2.0×10^{-10} to 7.2×10^{-10} over the charging window. The small coefficients are required because $\exp(22)$ is approximately 3.6×10^9. Even charging at 22 kW in all eight intervals costs only 11.90 reward units, compared with the shortage penalty of 100 € per missing kWh, ensuring charging is always better than not charging. Consequently, the original costs of 0, 5, and 11 kW are almost indistinguishable, while 22 kW constitutes a separate, much more expensive tier. Jumps between adjacent actions are reduced by the refined grid, allowing more graduated control, although the cost function remains nonlinear. The stylized model should therefore be interpreted as an artificial power-dependent reward rather than a realistic electricity tariff.

The EPEX environment uses the day-ahead price data described in Section 2, published one day before delivery and linearly interpolated within each hour to the eight quarter-hour charging decisions. The resulting series range from -0.026 € to 0.371 € per kWh, with a median of 0.110 € per kWh. Charging is billed at the realized price $\pi_t = \pi_t^{DA} + \varepsilon_t$, with $\varepsilon_t \sim \mathcal{N}(0, 0.015^2)$ €/kWh not observable before the decision; ε_t represents the gap between the day-ahead forecast and the delivery price and is simulated rather than drawn from real data.

6.3 Learning Algorithm and Benchmarks

Two Deep Q-Networks (DQN) are trained: a price-unaware agent in the stylized environment and an EPEX DQN in the EPEX environment. A multilayer perceptron with two hidden layers of 128 units and identical hyperparameters is used for both. Training lasts 200,000 time steps, equivalent to approximately 25,000 simulated charging days. Performance is evaluated on 1,000 paired episodes, using identical seeds so that capacity, initial SOC, demand, price day, and simulated price deviations match across strategies.

The operational benchmark is a price-blind greedy heuristic. Because demand is unknown, charging is not targeted at the realized requirement; instead, a fixed target $\tau = \min(C, \mu + 2\sigma) = \min(C, 40 \text{ kWh})$ is used, corresponding to an approximate 97.7% service level under the assumed normal distribution. At each step, the energy still missing from this target is spread over the remaining time, and the smallest sufficient charging power is selected.

An exact brute-force benchmark is used, evaluating every action sequence. For the original grid this requires $4^8 = 65{,}536$ sequences, evaluated on 100 episodes. The refined grid increases the search to $7^8 = 5{,}764{,}801$ sequences, so the EPEX oracle evaluation is restricted to 40 episodes. The realized demand and, in the EPEX setting, the realized billed prices, are known to this oracle benchmark in advance, making it a theoretical upper bound rather than an implementable policy.

6.4 Results

An episode succeeds when the final state of charge covers the realized demand ($SOC_8 \geq d$), so that no shortage penalty is added to r_8; success rate is therefore the primary reliability metric, alongside mean reward and, under EPEX prices, mean charging cost.

On the original four-action grid, the compressed cost tiers described above leave the price-unaware DQN with little reason to modulate power: it collapses onto a single action instead of spreading decisions the way greedy does, and it trails greedy on both success and reward. The refined seven-action grid restores graduated control and closes most of this gap (Table 18).

Under EPEX prices, with the refined grid, greedy, the price-unaware DQN, and the EPEX DQN are evaluated on the same paired episodes and billed at the same realized prices, so success rate, cost, and reward are directly comparable across policies in Table 19 (visualized in Figure 30); only the training objective differs between them. The EPEX DQN reaches the lowest cost by a wide margin, but its success rate is far below the other policies, so part of its saving reflects under-buying reserve rather than better scheduling. The price-unaware DQN nearly matches greedy's reliability but stays 18.8% more expensive on episodes where both succeed, because it was trained to minimize the stylized exponential cost rather than real per-kWh prices.

Restricted to the 752 episodes where both greedy and the EPEX DQN succeed, the EPEX DQN saves €0.63 per day (41.7%) on average, cheaper in the large majority of them and increasingly so as the day-ahead price spread within the window grows (Figure 31) – though this excludes the 24.8% of episodes it fails outright. The learned policies follow the intended qualitative pattern (Figure 32): power tracks state of charge and remaining time for the price-unaware DQN, while the EPEX DQN additionally defers to cheap intervals except when low state of charge makes charging urgent.

6.5 Operational Implications and Limitations

The price-unaware DQN's edge over greedy is an artifact of the artificial exponential reward: it evaporates under real EPEX billing, where a policy trained on the stylized cost turns out more expensive rather than cheaper (Table 19). Only the EPEX DQN, trained directly on real prices, delivers a genuine advantage.

That advantage is not free: at 75.2% success against greedy's 98.9%, part of the EPEX DQN's saving is forgone safety margin rather than pure price timing, and a policy that fails to fully charge one in four vehicles is not operationally viable on its own. Before deployment it would need an explicit safety layer – for example a hard minimum-SOC floor in the final intervals – to close this reliability gap without giving up much of the timing-based saving. That saving grows with day-ahead volatility (Figure 31), so RL-based charging is worthwhile only alongside a genuinely dynamic electricity tariff.

7 Conclusion

The analysis indicates that a potential electric-taxi operator should begin with a conservative, zone-based planning system rather than rely immediately on complex prediction and control models. Observed taxi demand is strongly concentrated: more than 70% of spatially assigned pickups originate in Near North Side, O'Hare, the Loop, and Near West Side, and demand peaks on weekdays in the late afternoon. The Gaussian Mixture Model confirms three broader operational hot spots—the extended city center, O'Hare, and Midway—whose relative importance changes over the day. These patterns support staging vehicles near central business and transport zones and rebalancing them toward airports during evening and overnight periods.

For forecasting, complexity did not consistently improve operational performance. The historical profile baseline remained difficult to outperform, and most pooled SVR and neural-network variants achieved negative demand-weighted skill. Only selected configurations, including the deep one-hour neural network and dedicated RBF models for high-demand areas, produced small improvements. A practical architecture should therefore retain the transparent profile baseline for most areas and reserve specialized models for a limited number of high-volume zones where forecast improvements have the greatest allocation value.

The charging experiment leads to a similar recommendation. Under EPEX pricing, the price-blind greedy policy achieved a 98.9% success rate and outperformed both DQN agents in mean reward after shortage penalties were included. Although the price-aware DQN reduced electricity costs, its 75.2% success rate is insufficient for fleet operations. Price-aware optimization should therefore operate only within hard minimum-SOC and departure-energy constraints. For vehicles returning regularly to a depot, private charging offers the greatest control over availability and dynamic tariffs; public fast charging remains useful as backup. The present data do not determine the optimal number or location of chargers because depot locations, grid limits, charger occupancy, and infrastructure costs are unavailable.

Overall, the project is a proof of concept rather than a deployment-ready system: its strengths are the large trip dataset, the explicit spatial and temporal resolution analysis, and operational baselines linking demand, forecasting, and charging, while its main limitations are anonymized centroid locations, incomplete Census Tract coverage, a threshold-dependent idle-time proxy, and simplified charging assumptions. The natural next step is an interpretable system combining demand profiles, hot-spot-specific forecasts, and rule-based safe charging, built out once complete vehicle duty cycles and charging-infrastructure data are available.

8 Appendix: Detailed Tables and Figures

Tables and figures below are referenced from the main text and grouped by chapter, in the same order as they are cited. All Predictive Analytics figures and score tables – the March-week model comparisons, the per-area skill maps, and the per-resolution score tables – are collected here and pointed to from the Results and Validation Strategy sections. The Reinforcement Learning environment assumptions, action-grid and EPEX comparisons, and policy-behavior figures are collected the same way and pointed to from the Problem Setup and Results sections.

8.1 Cleaned Dataset: Column Reference

Every column of `data/chicago_taxi_2024_2026_clean.parquet` (14,465,662 rows), with dtype, completeness, and distinct-value count as measured on that file, plus what the column means and any caveat carried over from the cleaning steps in Section 3 (e.g. census-tract masking, the two coordinate resolutions).

Table 1: Data dictionary, cleaned dataset.

Column	Dtype	Non-Null %	Distinct	Description
trip_id	str	100.0%	14,465,662	Unique trip identifier (hash). Deduplicated in cleaning step 1.
taxi_id	str	100.0%	3,354	Anonymized vehicle identifier (hash).
trip_start_timestamp	datetime64[us]	100.0%	84,660	Local Chicago date/time the trip started.
trip_end_timestamp	datetime64[us]	100.0%	84,664	Local Chicago date/time the trip ended.
trip_seconds	float32	100.0%	9,722	Trip duration in seconds (>= 60s, capped at the 99.9th percentile).
trip_miles	float32	100.0%	4,303	Trip distance in miles (> 0, capped at the 99.9th percentile).
pickup_census_tract	float32	44.6%	149	Pickup census tract (FIPS code). Missing for ~55% of trips – privacy masking of low-volume tracts.
dropoff_census_tract	float32	43.3%	175	Dropoff census tract (FIPS code). Same privacy masking as pickup.
pickup_community_area	Int64	98.2%	77	One of 77 official community areas for the pickup. Missing means the pickup was outside city limits.
dropoff_community_area	Int64	91.6%	77	Community area for the dropoff. Missing means the dropoff was outside city limits.
fare	float32	100.0%	9,078	Metered fare in USD (>= $3.25 flag-drop minimum).
tips	float32	100.0%	3,619	Tip amount in USD. 0 is valid – cash trips report no tip.
tolls	float32	100.0%	560	Toll charges in USD. 0 is valid – most trips use no toll roads.
extras	float32	100.0%	3,624	Additional surcharges in USD. 0 is valid.
trip_total	float32	100.0%	14,616	Total amount charged in USD (>= $3.25).
payment_type	category	100.0%	7	Payment method (Credit Card, Cash, Mobile, Prcard, Unknown, …). Left untouched by cleaning.
company	str	100.0%	45	Taxi affiliation / operating company. Left untouched by cleaning.
pickup_centroid_latitude	float32	98.2%	571	Pickup latitude: census tract centroid where known, else community area centroid (mixed resolution).
pickup_centroid_longitude	float32	98.2%	565	Pickup longitude, same mixed-resolution rule as latitude.

Column	Dtype	Non-Null %	Distinct	Description
dropoff_centroid_latitude	float32	92.1%	684	Dropoff latitude, same mixed-resolution rule.
dropoff_centroid_longitude	float32	92.1%	678	Dropoff longitude, same mixed-resolution rule.
pickup_flag	int64	100.0%	2	1 if pickup_community_area is present, 0 if missing (outside Chicago).
dropoff_flag	int64	100.0%	2	1 if dropoff_community_area is present, 0 if missing (outside Chicago).
pickup_ca_centroid_lat	float32	98.2%	77	Pickup community area centroid latitude – always CA-level resolution (added in step 3).
pickup_ca_centroid_lon	float32	98.2%	77	Pickup community area centroid longitude – always CA-level resolution.
dropoff_ca_centroid_lat	float32	91.6%	77	Dropoff community area centroid latitude – always CA-level resolution.
dropoff_ca_centroid_lon	float32	91.6%	77	Dropoff community area centroid longitude – always CA-level resolution.

8.2 Descriptive Analytics: Temporal and Trip Patterns

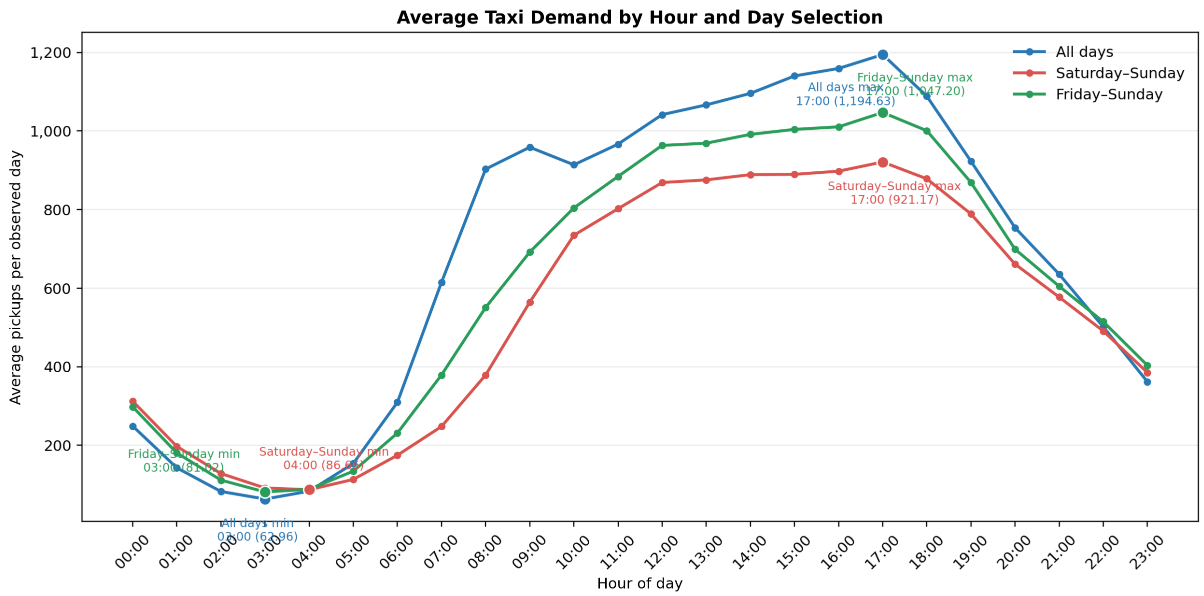

Figure 1: Comparison of average hourly pickup demand for all days, the strict Saturday–Sunday weekend, and the extended Friday–Sunday period. Highlighted points and labels show the minimum and maximum of each profile.

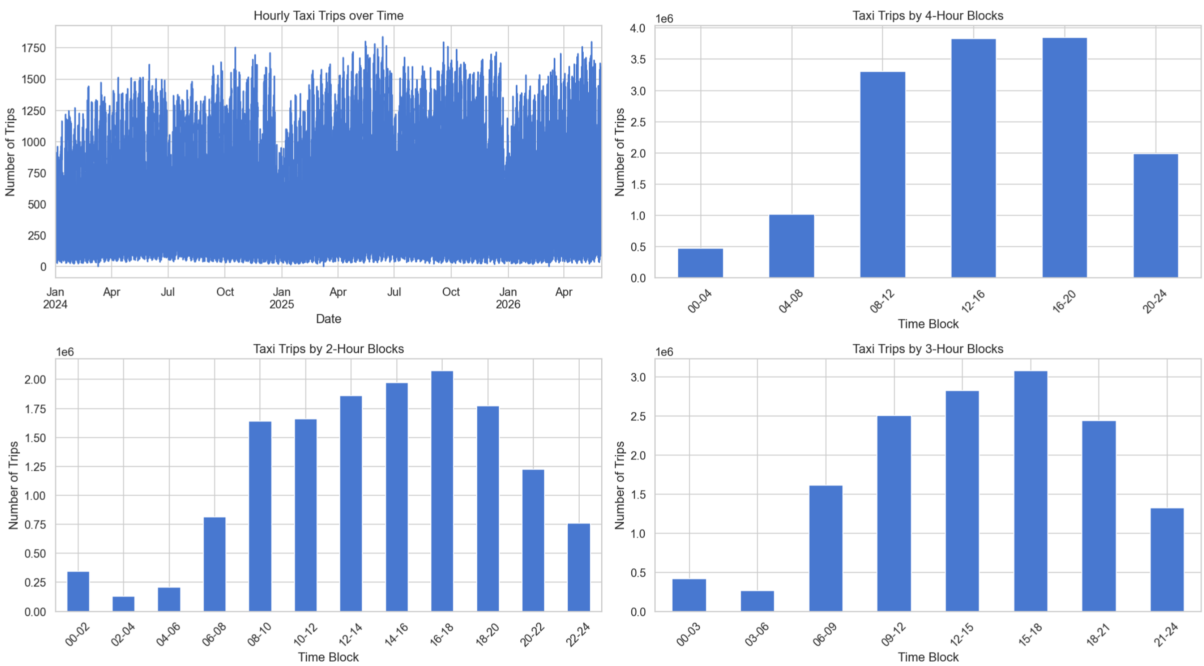

Figure 2: Taxi demand at hourly, two-hour, three-hour, and four-hour temporal resolutions. Wider bins smooth short-lived variation into broader operational periods.

Table 2: Average pickup demand by weekday.

Day	Average pickups per observed day
Monday	16,382.15
Tuesday	17,681.13
Wednesday	18,358.35
Thursday	18,856.44
Friday	17,623.75
Saturday	13,313.98
Sunday	12,591.04

Table 3: Distribution of trip characteristics.

Measure	Median	Mean
Trip distance	3.85 miles	7.09 miles
Trip duration	16.00 minutes	20.97 minutes
Metered fare	USD 15.25	USD 22.06
Total charge	USD 17.75	USD 27.01

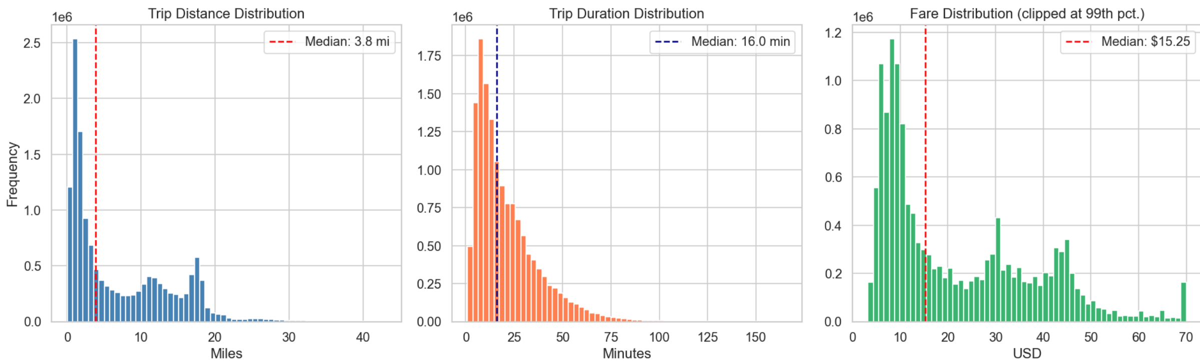

Figure 3: Distributions of trip distance, trip duration, and metered fare. Fare is clipped at the 99th percentile for visualization only; summary statistics use the full cleaned data.

	trip_miles	trip_minutes	fare	trip_total
count	14465662.00	14465662.00	14465662.00	14465662.00
mean	7.09	20.97	22.06	27.01
std	6.84	16.36	16.84	23.42
min	0.01	1.00	3.25	3.25
25%	1.40	9.00	8.50	10.50
50%	3.85	16.00	15.25	17.75
75%	12.20	28.35	33.50	40.50
max	43.03	165.37	6950.69	8940.88

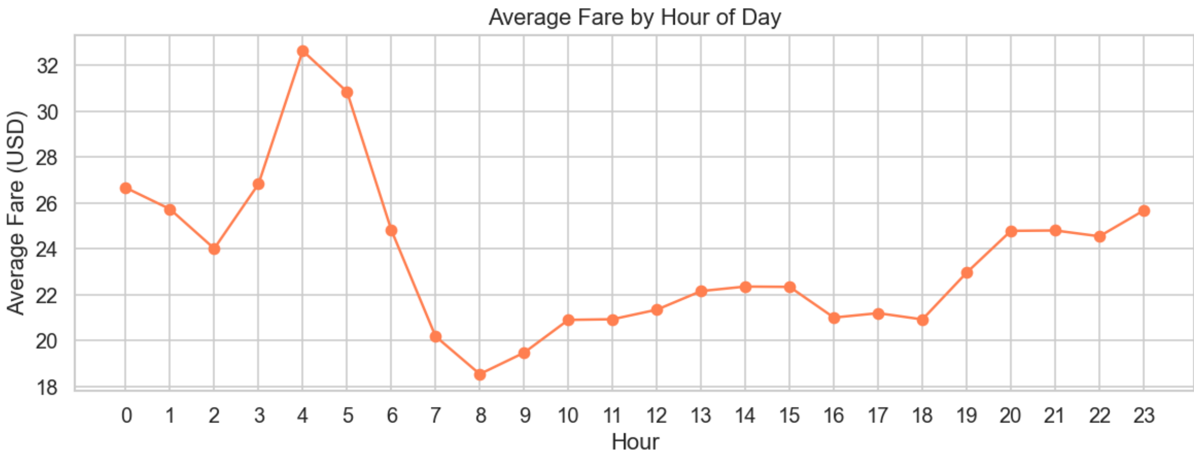

Figure 4: Mean metered fare by hour of day.

Table 4: Sensitivity of the consecutive-trip idle-time proxy.

Maximum retained gap	Transitions retained	Share of non-negative gaps	Mean proxy	Median proxy
1 hour	9,366,946	65.77%	21.20 min	15.00 min
2 hours	10,986,502	77.14%	31.88 min	15.00 min
4 hours	**12,072,245**	**84.76%**	**44.49 min**	**30.00 min**
6 hours	12,297,062	86.34%	49.09 min	30.00 min
8 hours	12,395,975	87.03%	52.08 min	30.00 min

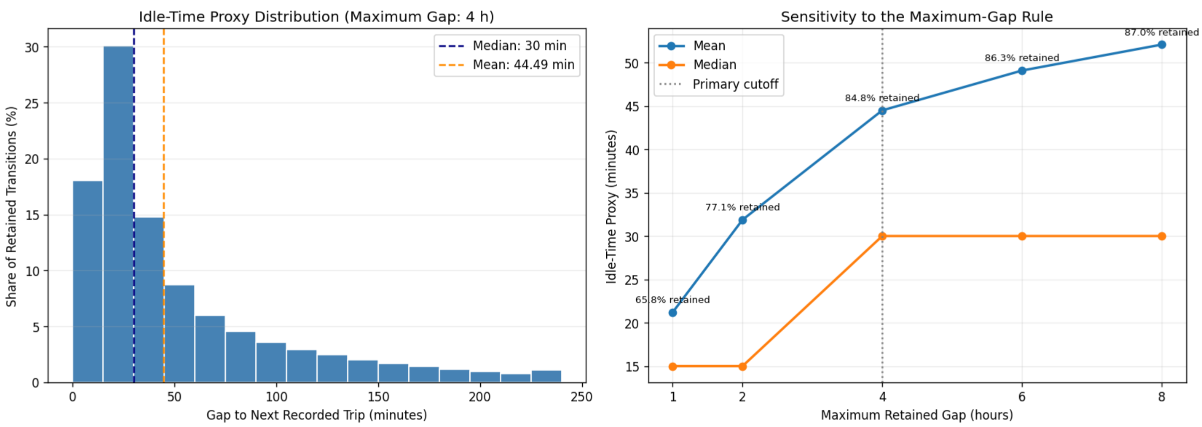

Figure 5: Idle-time proxy distribution under the four-hour rule (left) and sensitivity to the maximum retained gap (right). Percent labels show the retained share of non-negative consecutive-trip gaps.

8.3 Descriptive Spatial Analytics: Tables and Hot Spots

Table 5: Pickup and drop-off volumes for the four highest-volume pickup Community Areas.

Community Area	Pickup trips	Drop-off trips
Near North Side	3,264,625	3,528,594
O'Hare	2,985,521	579,319
Loop	2,374,677	2,452,298
Near West Side	1,467,354	1,450,354

Table 6: Dominant Community Area by four-hour block. Pickups use start time and drop-offs use end time.

Time block	Leading pickup area	Pickup trips	Leading drop-off area	Drop-off trips
00:00–04:00	O'Hare	168,417	Near North Side	107,260
04:00–08:00	Near North Side	210,845	Loop	180,906
08:00–12:00	Near North Side	731,045	Near North Side	763,901
12:00–16:00	Near North Side	894,316	Near North Side	945,428
16:00–20:00	Near North Side	926,865	Near North Side	1,031,329
20:00–24:00	O'Hare	703,740	Near North Side	529,528

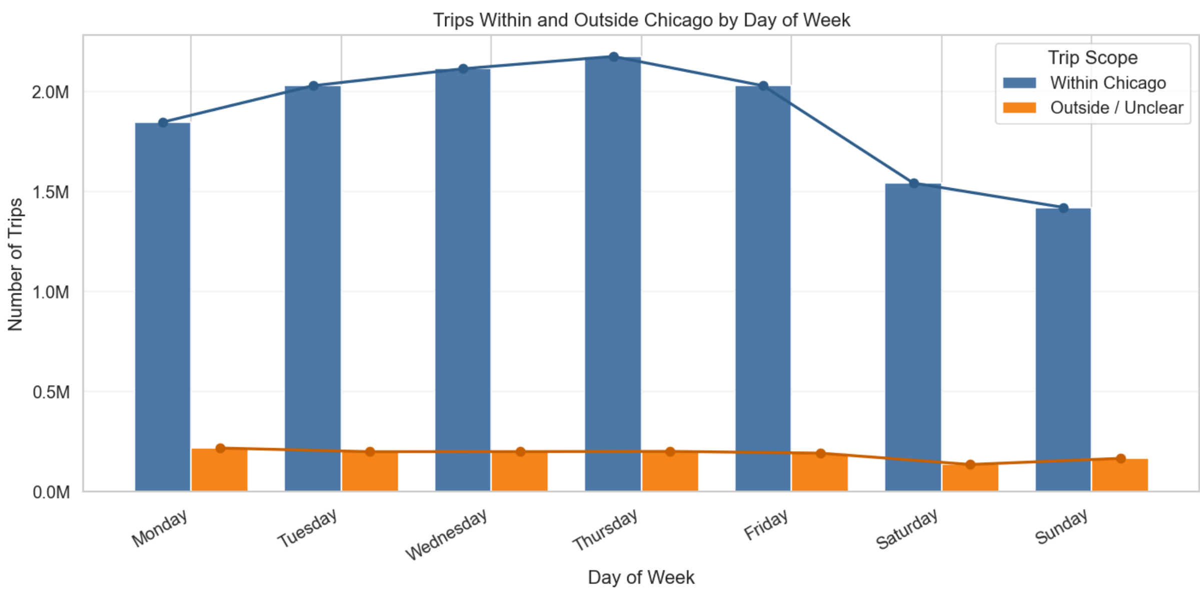

Figure 6: Trips with both pickup and drop-off assigned to a Chicago Community Area, compared with trips having at least one outside or unclear assignment, by weekday.

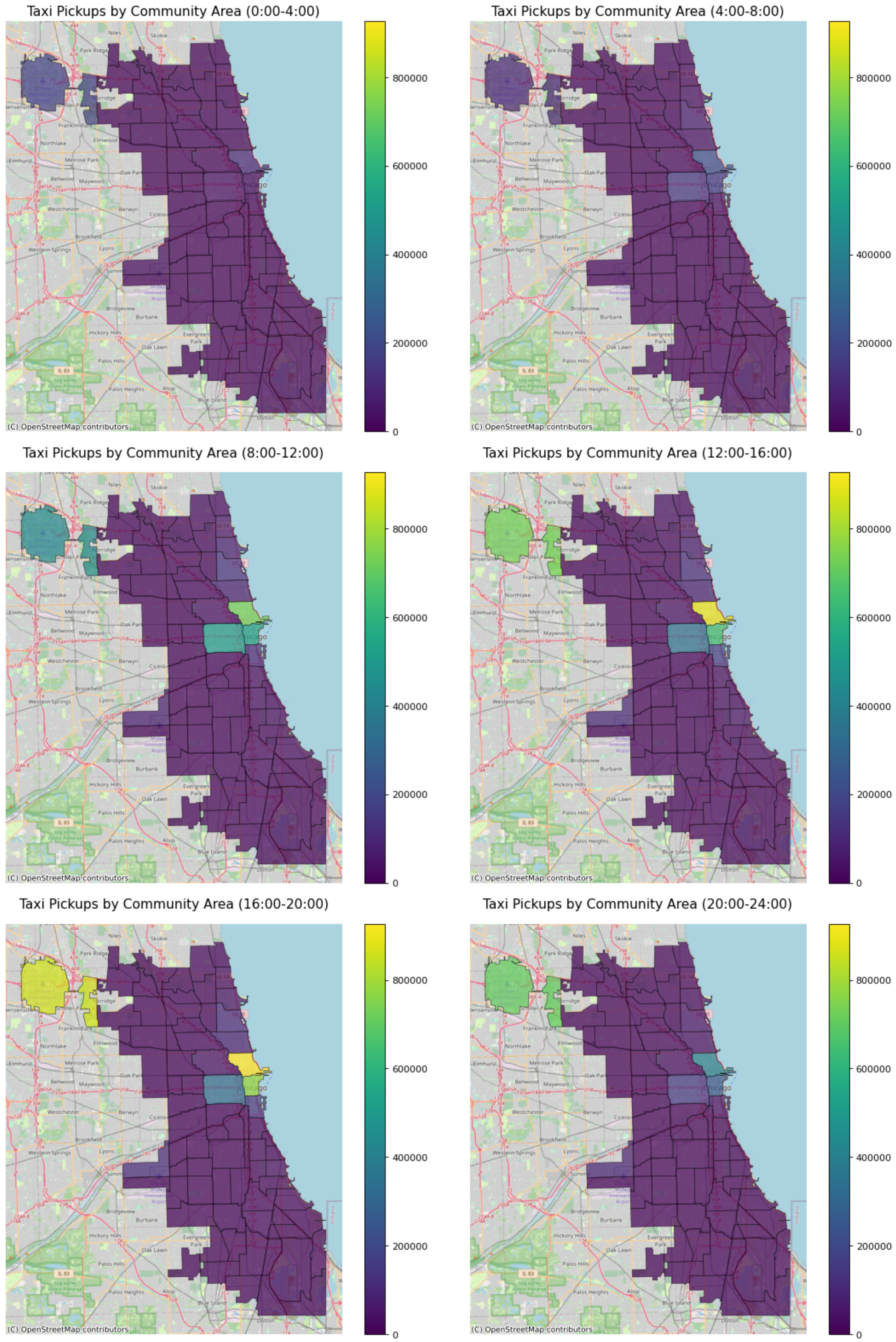

Figure 7: Pickup demand by Community Area and four-hour time block.

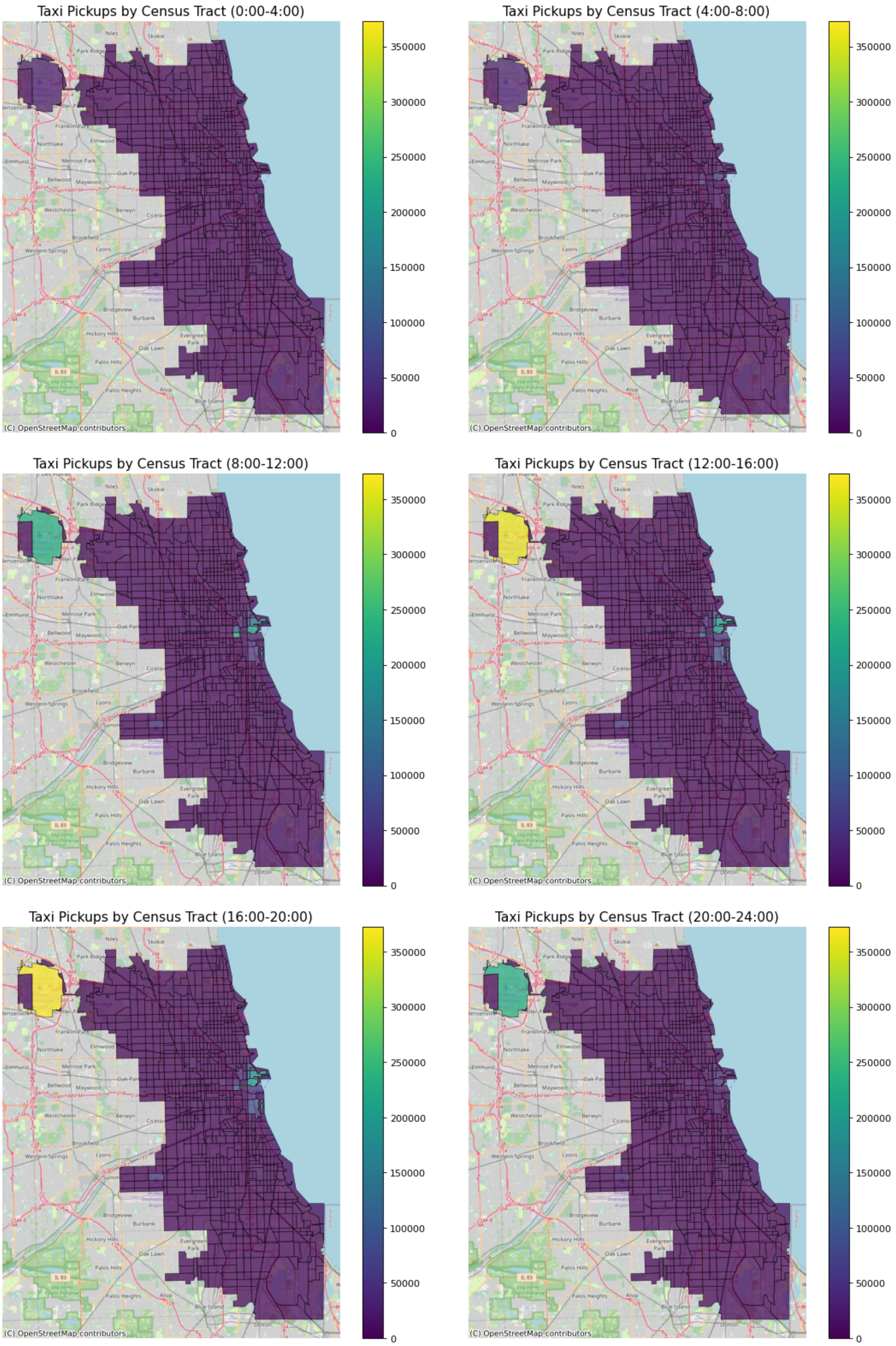

Figure 8: Pickup demand by Census Tract and four-hour time block. Maps use only trips for which both Census Tract identifiers are available.

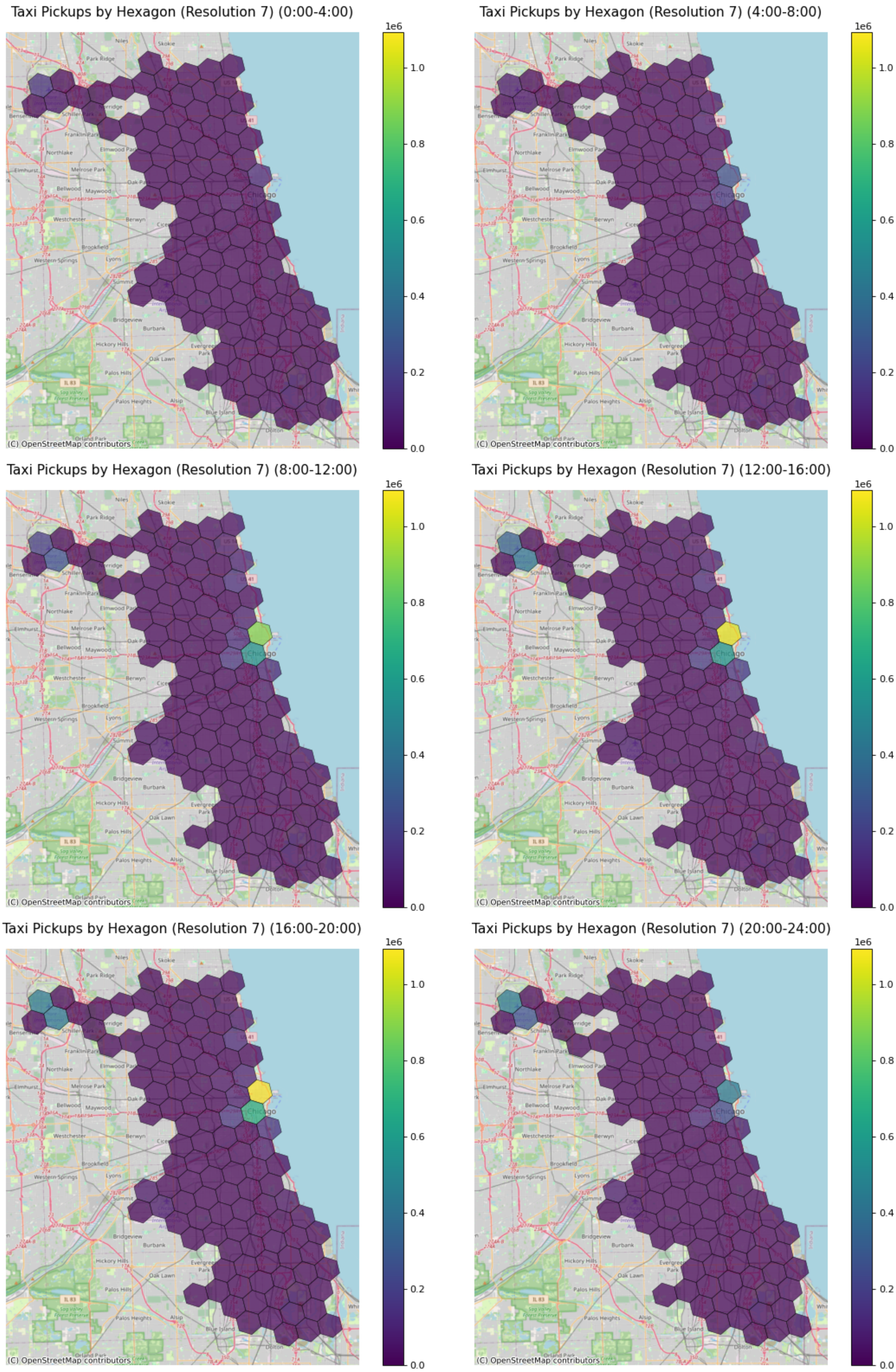

Figure 9: Pickup demand by H3 resolution-7 cell and four-hour trip-start block.

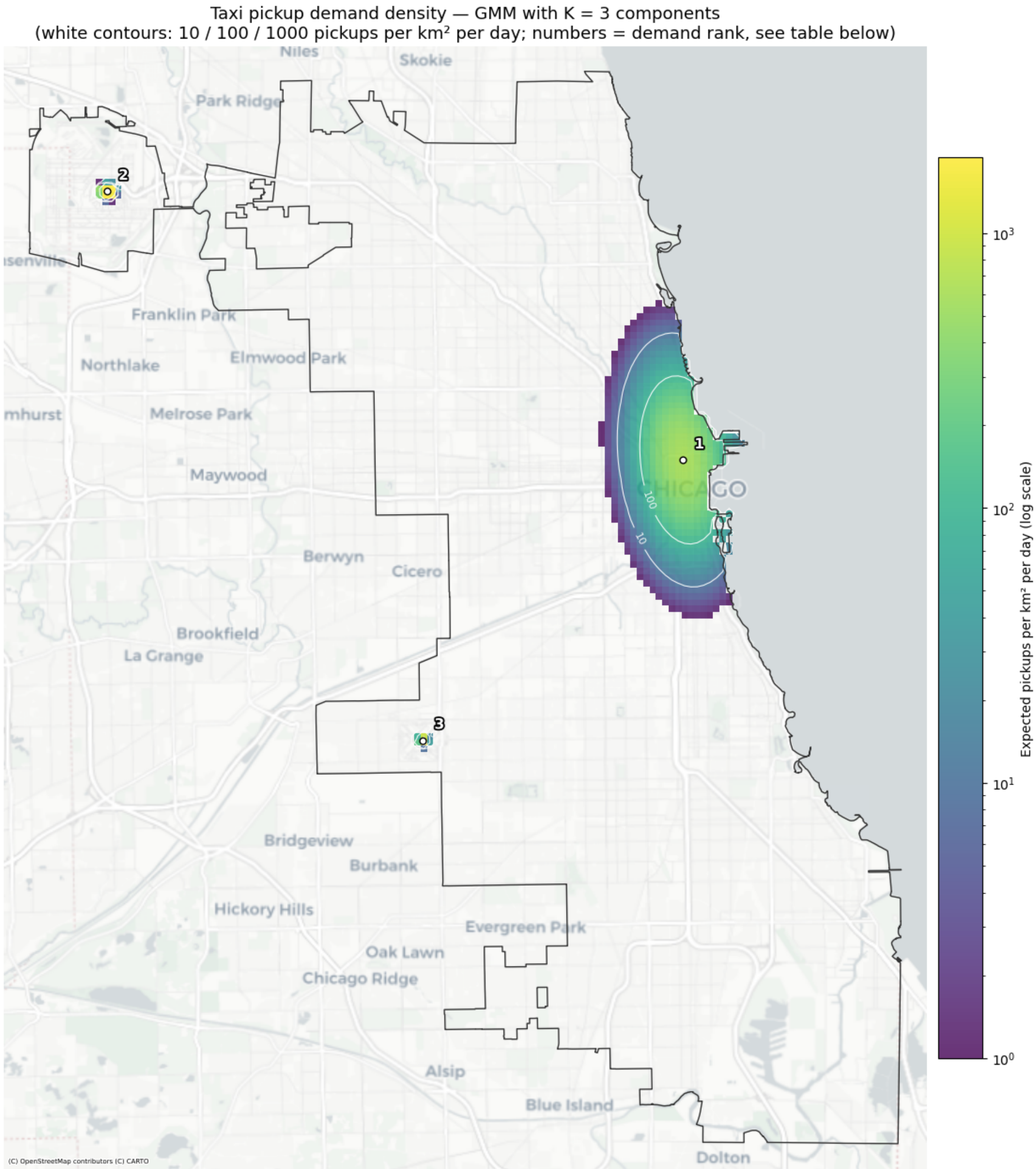

Figure 10: Fitted GMM demand density over the city, in expected pickups per km² per day (log scale). Numbered markers are the three components' centers, ranked by demand share; white contours mark the 10 / 100 / 1000 pickups per km² per day levels.

Table 7: Final three-component GMM hot spots.

Rank	Operational hot spot	Demand share	Tract-resolved pickups/day	Spatial extent (σ major/minor)
1	Extended city center (center in Loop)	76.3%	5,830	1,746 m / 878 m
2	O'Hare	21.0%	1,607	123 m / 100 m

Rank	Operational hot spot	Demand share	Tract-resolved pickups/day	Spatial extent (σ major/minor)
3	Midway/Garfield Ridge	2.6%	201	101 m / 100 m

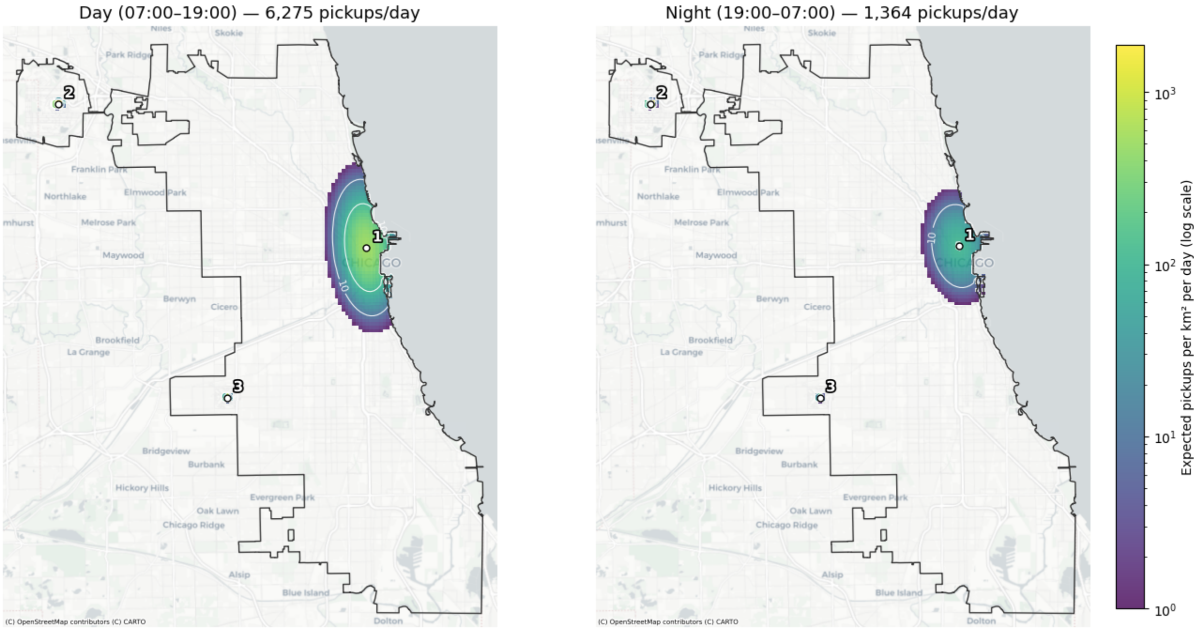

Figure 11: Three-component GMM hot spots fitted separately for daytime (07:00–19:00) and nighttime (19:00–07:00). Percentages show each component's share of tract-resolved pickup demand within the respective period.

```
Day (07:00-19:00): LOOP (80 %), OHARE (18 %), GARFIELD RIDGE (2 %)
Night (19:00-07:00): LOOP (61 %), OHARE (36 %), GARFIELD RIDGE (4 %)
```

8.4 Predictive Analytics: Baseline

The **profile baseline** predicts each unit's demand from its own historical calendar profile. Let $b(t) = (\text{hour}(t), \text{dow}(t), \text{month}(t))$ be the calendar bucket of time window t – hour-of-day, day-of-week (0–6), and month. For a unit i and a test window t, the baseline predicts the mean training-set demand of that unit over all training windows falling in the same bucket,

$$\hat{y}_{i,t}^{\text{base}} = \frac{1}{|\mathcal{T}_i^{\,b(t)}|} \sum_{s \in \mathcal{T}_i^{\,b(t)}} y_{i,s}, \qquad \mathcal{T}_i^{\,b} = \{\, s \in \text{train} : b(s) = b \,\}, \tag{1}$$

computed independently for every unit i, so each unit gets its own reference. At 4-hour resolution the hour component takes the window's start hour $(0, 4, \dots, 20)$, giving $6 \times 7 \times 12 = 504$ buckets per unit; at 1-hour resolution there are $24 \times 7 \times 12 = 2016$. The implementation falls back to unit i's overall training-set mean $\bar{y}_i^{\text{train}}$ for any bucket absent from training, but with two full years (2024–2025) as the training window and a complete zero-filled grid, every bucket is populated, so this fallback never triggers at the resolutions used here.

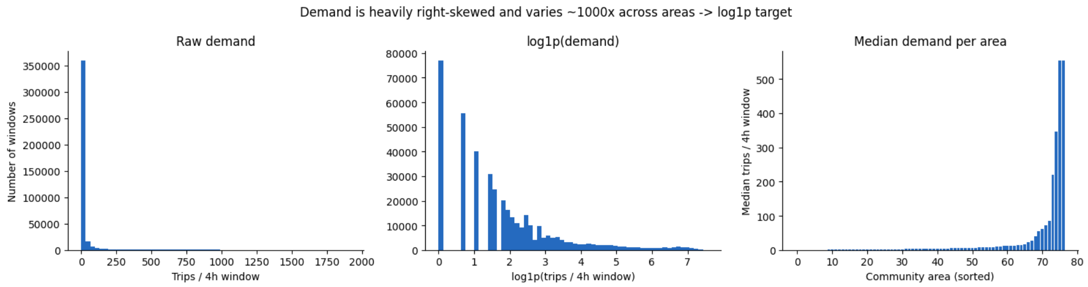

Figure 12: Raw per-window demand is heavily right-skewed, which is why both SVR and NN regress on `log1p(count)` rather than raw counts.

Table 8: Profile baseline test scores by resolution.

Resolution	Units	Pooled R^2	Weighted RMSE	Median per-unit R^2	Units with $R^2 < 0$
Community area, 4-hour	77	0.933	139.02	0.208	21 of 77
Community area, 1-hour	77	0.918	39.46	0.028	36 of 77
Census tract, 4-hour	179	0.896	100.99	0.195	56 of 179

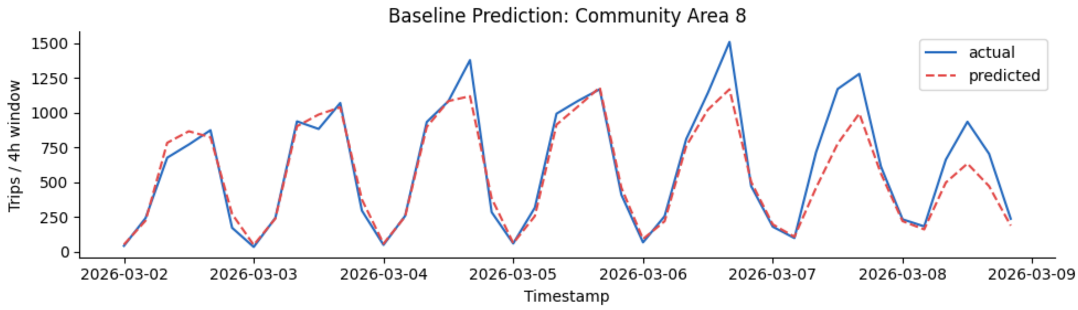

Figure 13: Profile baseline predictions vs. actual demand across the March evaluation week, highest-demand community area (community area, 4-hour resolution). The baseline tracks the daily and weekly cycle closely.

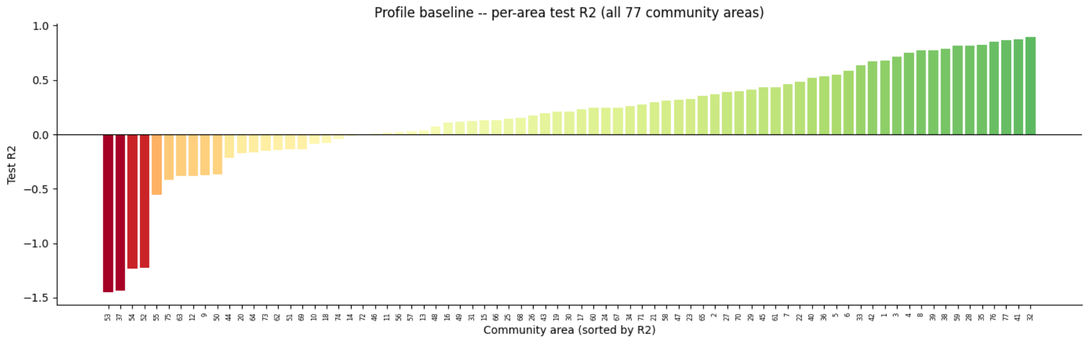

Figure 14: Profile baseline test R^2 per community area, 4-hour resolution. The pooled R^2 (0.933) is dominated by the highest-volume areas; the per-area distribution is much wider, with 21 of 77 areas negative.

8.5 Predictive Analytics: Metric Definitions

The two headline metrics used throughout the Predictive Analytics chapter are defined below. The per-unit **skill score** compares a model's mean absolute error to the profile baseline's for the same spatial unit i,

$$\text{skill}_i = 1 - \frac{\text{MAE}_{\text{model},\, i}}{\text{MAE}_{\text{baseline},\, i}} \tag{2}$$

so that 0 means tied with the baseline, 1 means zero error, and a negative value means worse than the baseline. The reported **demand-weighted skill** aggregates the per-unit skills with weights equal to each unit's median demand m_i, so the highest-volume units dominate,

$$\text{skill}_{\text{demand-weighted}} = \frac{\sum_i m_i \, \text{skill}_i}{\sum_i m_i}. \tag{3}$$

The **weighted RMSE** is a pooled, absolute-scale error over every test-set row j (a unit crossed with a time window), with y_j and $\hat{y}_j$ the actual and predicted demand and $\bar{d}_{u(j)}$ the train-period mean demand of the unit that row j belongs to,

$$\text{weighted RMSE} = \sqrt{\frac{\sum_j \bar{d}_{u(j)} \left(y_j - \hat{y}_j\right)^2}{\sum_j \bar{d}_{u(j)}}}. \tag{4}$$

8.6 Predictive Analytics: Community Area, 4-Hour Resolution

Table 9: SVR test scores, community area, 4-hour resolution. Per-area models cover only the 20 areas above the demand threshold.

Model	Units	Dem.-weighted skill	Median skill	Weighted RMSE	Pooled R^2
Profile baseline	77	0.000	0.000	139.02	0.933
LinearSVR, pooled	77	-0.827	-0.017	254.94	0.778
LinearSVR, demand-weighted	77	-0.675	-0.052	231.91	0.814
LinearSVR, demand-tiered	77	-0.732	0.000	243.60	0.800
RBF, tuned (10k subsample)	77	-0.232	-0.027	176.93	0.892
Polynomial, tuned (10k subsample)	77	-0.221	-0.049	171.71	0.896
LinearSVR, per-area	20	-0.112	-0.091	156.93	0.905
LinearSVR, per-area, tuned	20	-0.125	-0.116	161.10	0.901
RBF, per-area	20	-0.135	-0.105	163.37	0.897
RBF, per-area, tuned	20	**+0.012**	-0.002	143.80	0.921

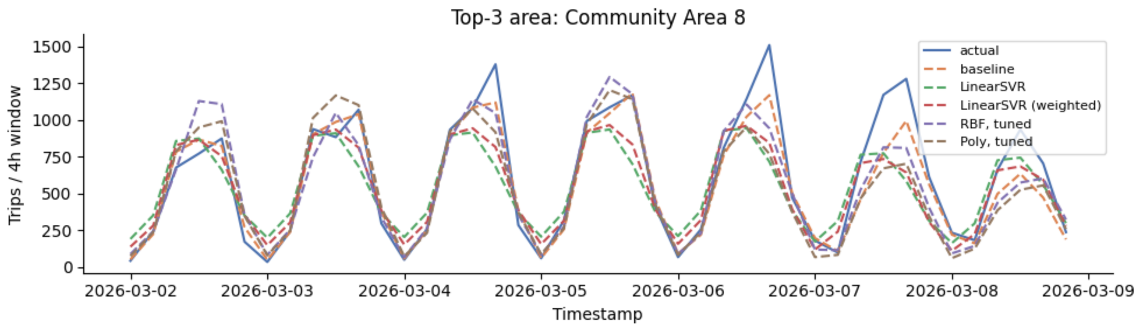

Figure 15: Actual, baseline and SVR-variant demand across the March evaluation week, Community Area 8, community area 4-hour resolution.

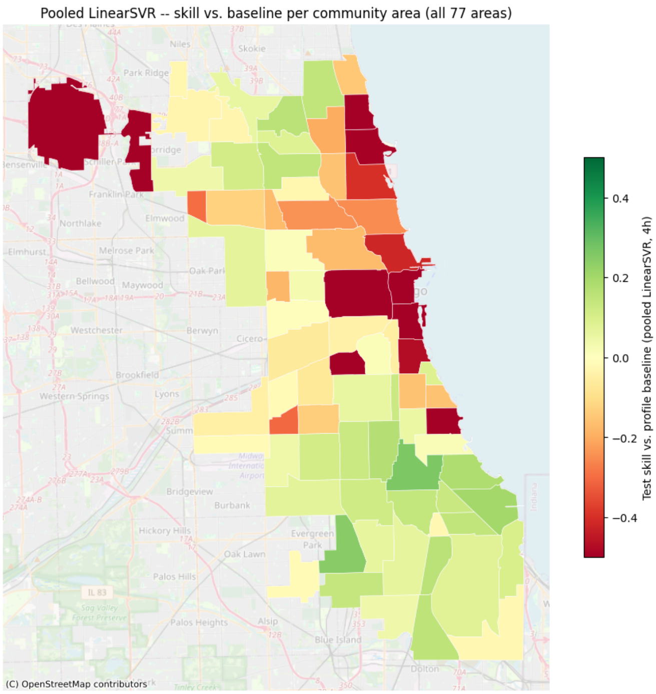

Figure 16: Pooled `LinearSVR` per-area skill vs. the profile baseline, all 77 community areas, 4-hour resolution.

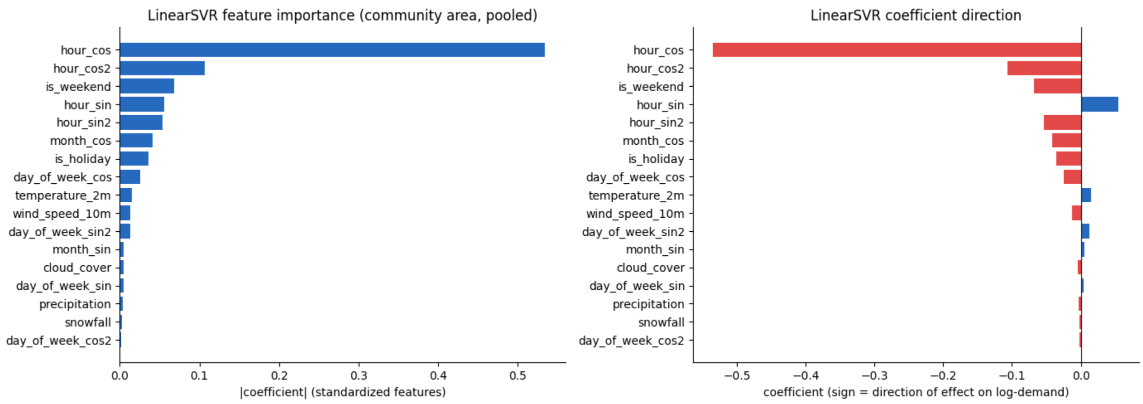

Figure 17: Pooled `LinearSVR` coefficients on the standardized features (community area, 4-hour). Left: magnitude (feature importance); right: sign. The calendar features – the hour-of-day harmonics above all, then weekend/holiday, weekday and month – dominate, while the weather variables rank lowest, with precipitation and snowfall near zero.

Table 10: Neural network test scores, community area, 4-hour resolution.

Model	Dem.-weighted skill	Median skill	Weighted RMSE	Pooled R^2	Fit time
Profile baseline	0.000	0.000	139.02	0.933	1s
Shallow NN (untuned)	-0.028	+0.097	144.96	0.925	324s
Deep NN (tuned)	-0.027	+0.071	145.93	0.927	2,333s

Model	Dem.-weighted skill	Median skill	Weighted RMSE	Pooled R^2	Fit time

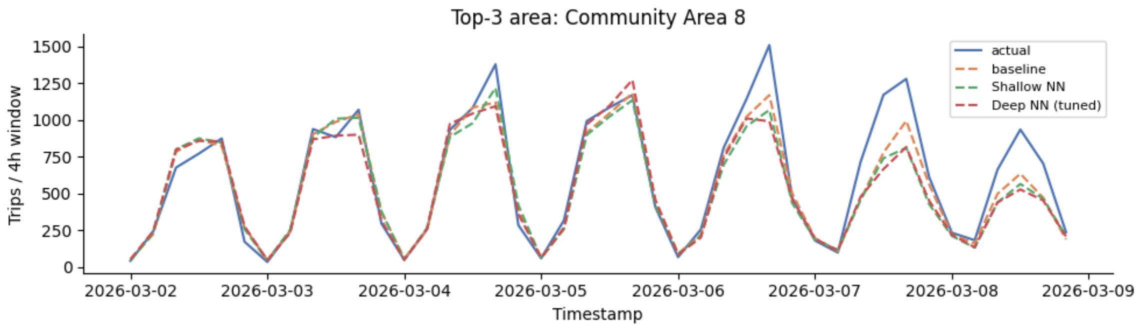

Figure 18: Actual, baseline and neural-network demand across the March evaluation week, Community Area 8 (highest demand), community area 4-hour resolution.

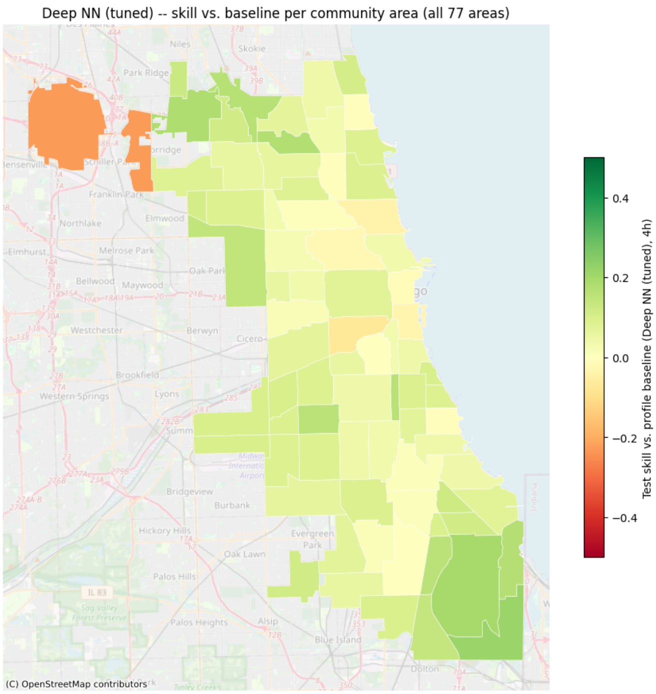

Figure 19: Tuned deep NN per-area skill vs. the profile baseline, all 77 community areas, 4-hour resolution. O'Hare (upper left) is the one clearly negative area.

8.7 Predictive Analytics: SVR Hyperparameter Grids

The kernel SVR configurations were tuned with `GridSearchCV`, scoring on R^2. Two searches were run: a **pooled** search over the full 77-area panel, tuned on a 10,000-row subsample with `KFold(3, shuffled)` cross-validation (kernel SVR training is quadratic-to-cubic in rows, so the full panel is infeasible); and a **per-area** search that fits a dedicated model for each of the 20 areas above the median-demand threshold, using `TimeSeriesSplit(3)`. The pooled grids are identical at 4-hour and 1-hour resolution; the per-area search was run only at 4-hour resolution.

Table 11: SVR grid-search spaces. Unless it is itself a grid dimension, ϵ is fixed at 0.1.

Search	Kernel	Grid searched	Cross-validation
Pooled (10k subsample)	Polynomial	$C \in \{1, 10, 100\}$, degree $\in \{2, 3\}$, gamma = scale	KFold(3, shuffled)
Pooled (10k subsample)	RBF	$C \in \{10, 100\}$, gamma $\in \{\text{scale}, 0.01\}$	KFold(3, shuffled)
Per-area (20 areas)	Linear	$C \in \{0.01, 0.1, 1, 10, 100\}$, $\epsilon \in \{0.01, 0.1\}$	TimeSeriesSplit(3)
Per-area (20 areas)	RBF	$C \in \{10, 100, 1000\}$, gamma $\in \{\text{scale}, 0.01\}$	TimeSeriesSplit(3)

8.8 Predictive Analytics: Neural Network Architecture Search

The deep network was selected by a **one-factor-at-a-time (OFAT) screen**: starting from the reference configuration below, each candidate turns exactly one knob so its effect is isolated against a common baseline. Every knob that improved demand-weighted skill over the reference was then combined into a single config (with the mutually exclusive capacity settings reduced to the single best-improving one), and the reported deep network is whichever of the reference, the single-knob variants, or the combined config scored best. This search was run once at the default community-area, 4-hour resolution and the selected config was reused unchanged at the 1-hour and census-tract settings.

Table 12: Neural network hyperparameter search space. All candidates use the Adam optimizer and ReLU activations, and are trained for up to 200 epochs with early stopping (patience 8) on count-space demand-weighted RMSE.

Hyperparameter	Reference	Values screened (one at a time)
Width (2 hidden layers)	128 / 64	64 / 32, 256 / 128, 512 / 256
Depth	2 layers (128 / 64)	5 layers (512 / 256 / 128 / 64 / 32)
Dropout	0.0	0.2
Weight decay (L2)	0.0	1e-4
Learning rate	1e-3	1e-4, 3e-4, 3e-3, 1e-2
Batch size	512	128, 256, 1024, 2048

8.9 Predictive Analytics: O'Hare Time-Feature Experiment

To test whether the pooled network could be nudged toward O'Hare's atypical round-the-clock demand, O'Hare-specific time features were added – the area-76 dummy interacted with the hour-of-day sine/cosine harmonics – on top of the standard feature set, and the shallow and deep networks were refit on the same panel, baseline, and methodology as Table 10. The features did not help. The best model from the experiment, the shallow network, reached demand-weighted skill −0.030 – no better than the −0.028 it scores without them (Table 10) – and the deep network was substantially worse.

Table 13: Neural network test scores with added O'Hare-specific time features, community area, 4-hour resolution. Compare the base-feature versions in Table 10 (shallow −0.028, deep −0.027).

Model	Dem.-weighted skill	Median skill	Weighted RMSE	Pooled R^2
Profile baseline	0.000	0.000	139.02	0.933
Shallow NN (+ O'Hare time features)	-0.030	+0.097	144.38	0.926
Deep NN (+ O'Hare time features)	-0.186	+0.071	166.64	0.906

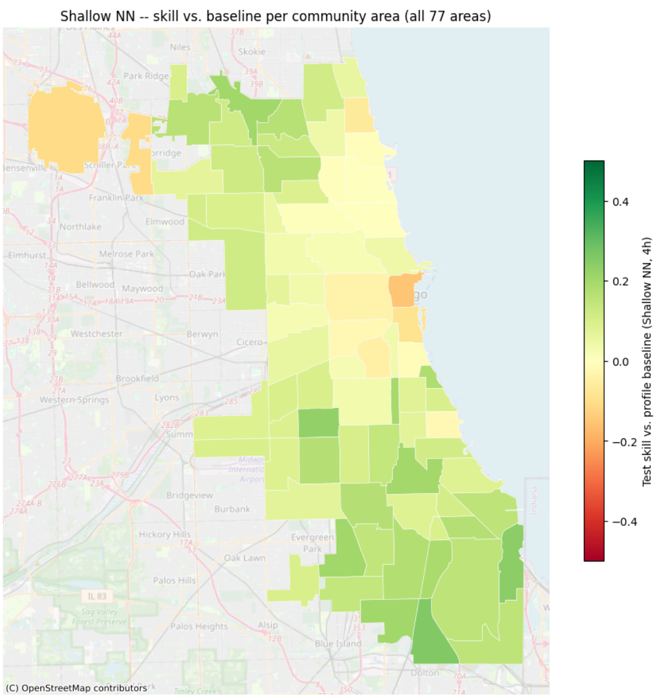

Figure 20: Shallow network (with added O'Hare-specific time features) per-area skill vs. the profile baseline, all 77 community areas, 4-hour resolution – the best-scoring model from the experiment. O'Hare (upper left) remains among the weaker areas, and the added features leave overall skill essentially unchanged.

8.10 Predictive Analytics: Community Area, 1-Hour Resolution

Table 14: SVR test scores, community area, 1-hour resolution.

Model	Dem.-weighted skill	Median skill	Weighted RMSE	Pooled R^2
Profile baseline	0.000	0.000	39.46	0.918
LinearSVR, pooled	-1.016	+0.022	76.96	0.687
RBF, tuned	-0.331	+0.015	54.53	0.835
Polynomial, tuned	-0.361	0.000	56.14	0.793

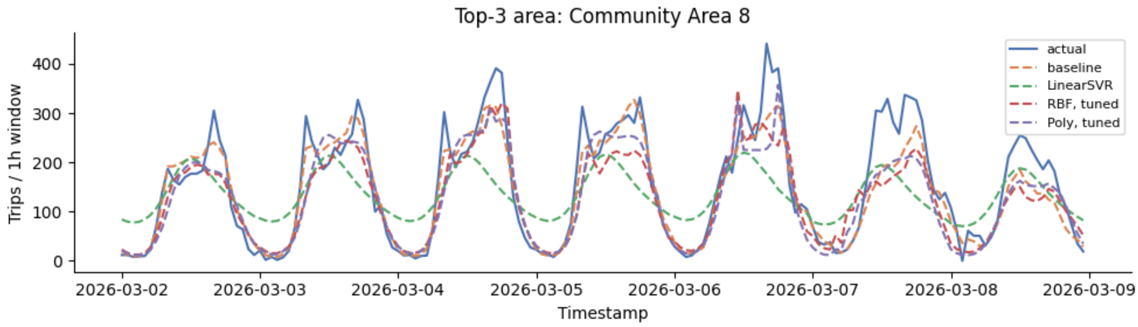

Figure 21: SVR March-week demand, Community Area 8, 1-hour resolution.

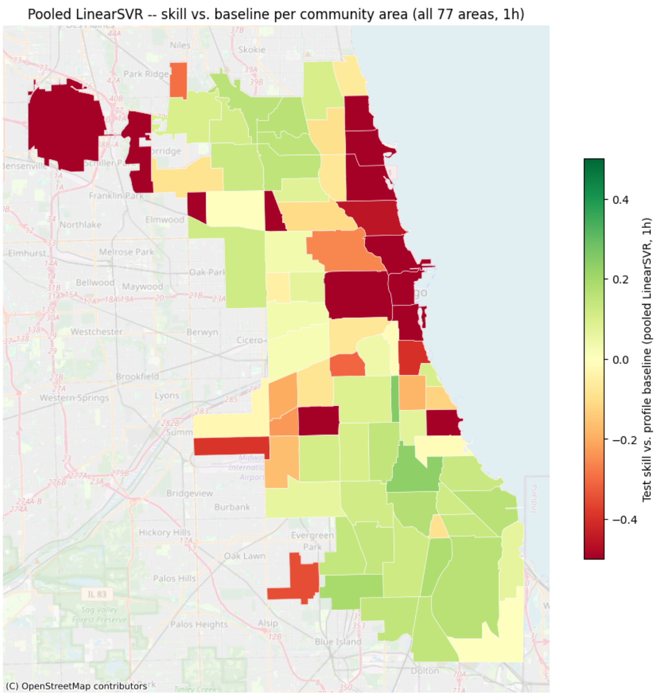

Figure 22: Pooled `LinearSVR` per-area skill vs. the profile baseline, all 77 community areas, 1-hour resolution.

Table 15: Neural network test scores, community area, 1-hour resolution.

Model	Dem.-weighted skill	Median skill	Weighted RMSE	Pooled R^2
Profile baseline	0.000	0.000	39.46	0.918
Shallow NN (untuned)	-0.048	+0.118	42.28	0.903
Deep NN (tuned)	**+0.062**	+0.090	36.26	0.928

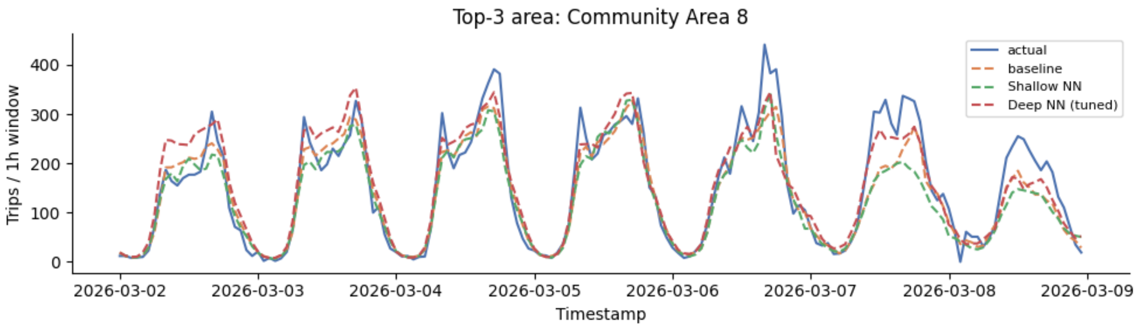

Figure 23: Neural-network March-week demand, Community Area 8, 1-hour resolution.

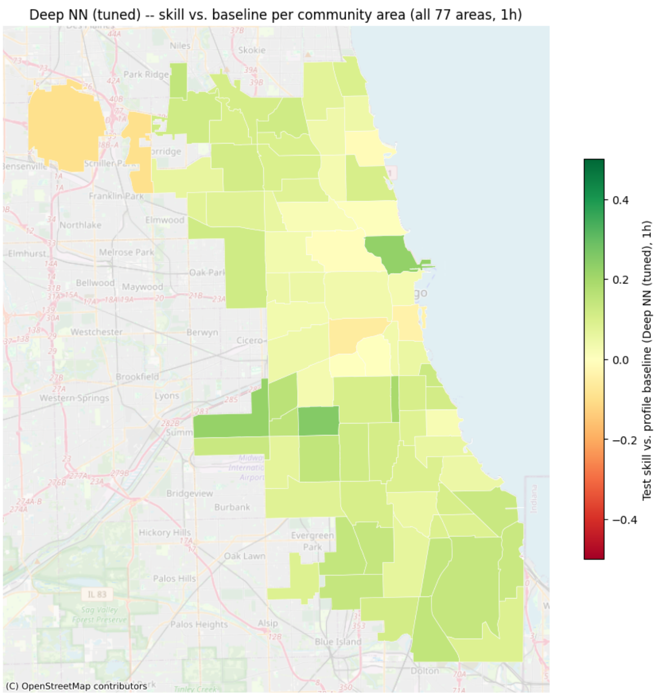

Figure 24: Tuned deep NN per-area skill vs. the profile baseline, all 77 community areas, 1-hour resolution.

8.11 Predictive Analytics: Census Tract, 4-Hour Resolution

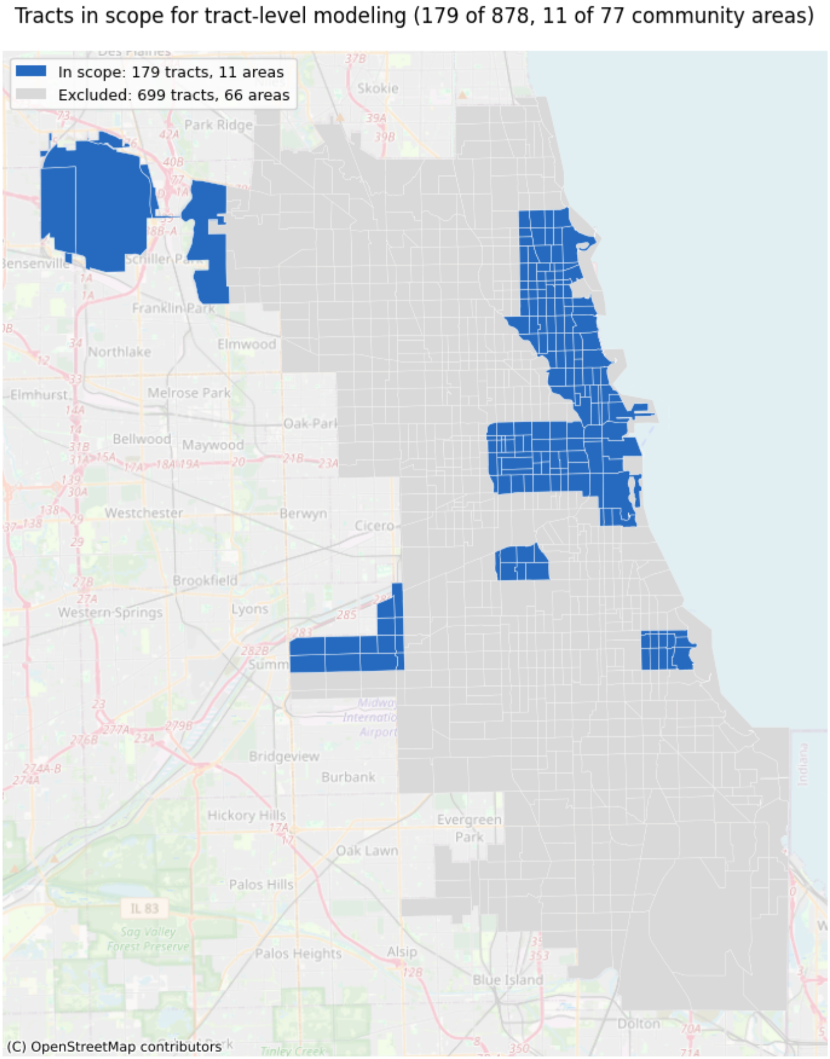

Figure 25: Scope map: the 179 of 878 census tracts with enough tract-precision data to support a tract-level model, across 11 community areas.

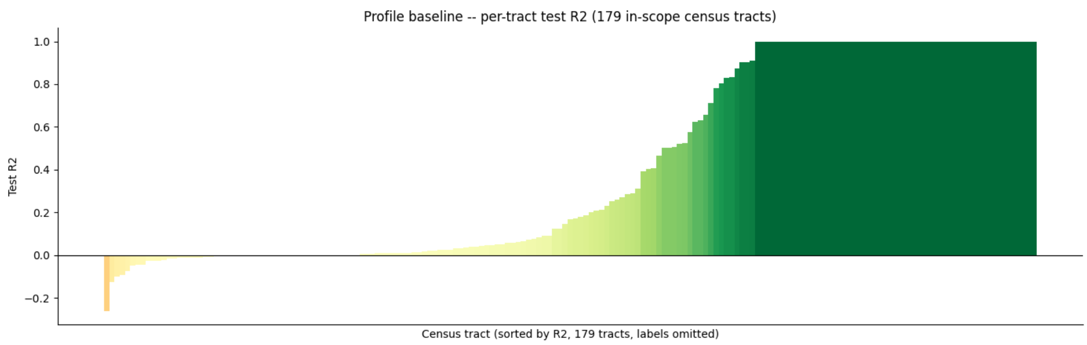

Figure 26: Profile baseline test R^2 per census tract, 4-hour resolution. Wider spread than the community-area version (Figure 14), consistent with more units competing for the same total demand.

Table 16: Neural network test scores, census tract, 4-hour resolution (179 in-scope tracts).

Model	Dem.-weighted skill	Weighted RMSE	Pooled R^2
Profile baseline	0.000	100.99	0.896
Shallow NN (untuned)	-0.098	113.77	0.868
Deep NN (reused 4h config)	-0.340	138.71	0.821

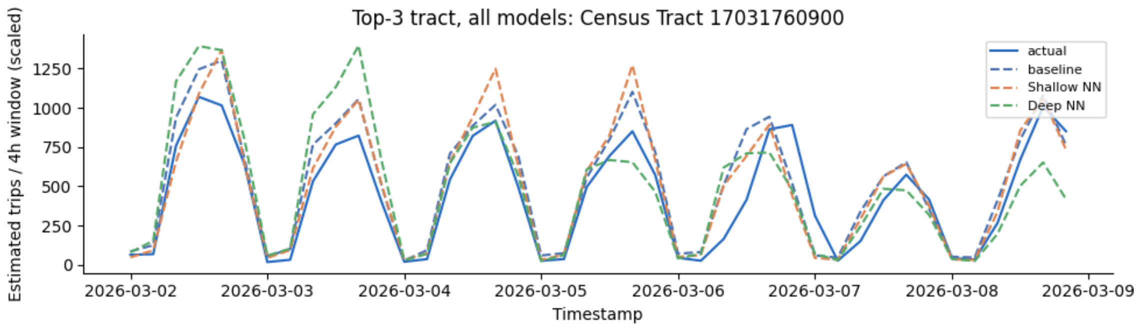

Figure 27: Actual, baseline and neural-network demand across the March evaluation week, top-demand census tract (17031760900), 4-hour resolution (estimated trips after inverse-retention scaling).

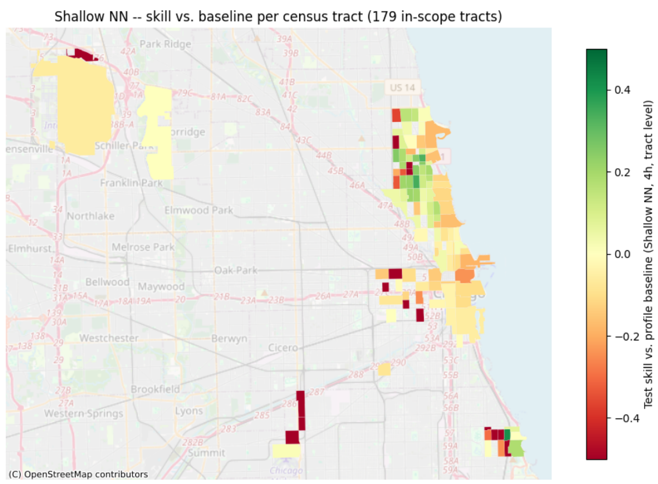

Figure 28: Tuned deep NN per-tract skill vs. the profile baseline, 179 in-scope census tracts, 4-hour resolution.

8.12 Predictive Analytics: Year-over-Year Demand Growth

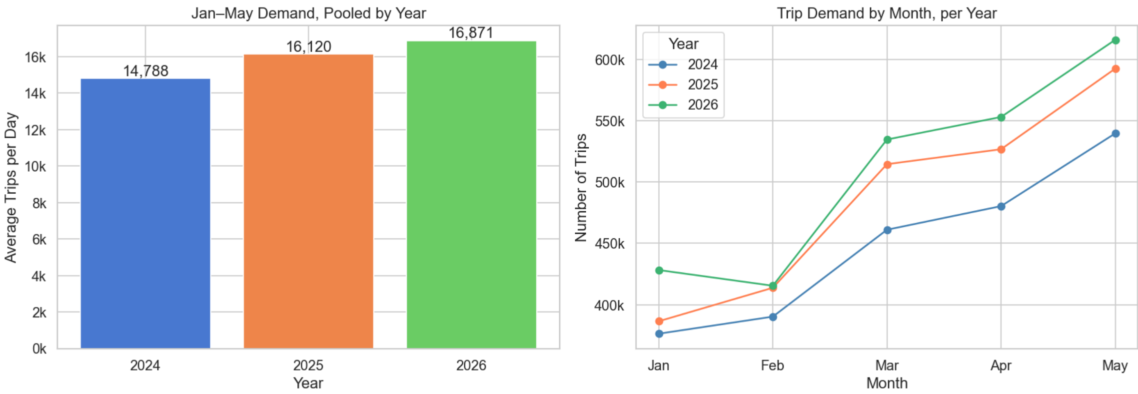

Figure 29: Year-over-year demand growth over the January–May window common to all three years (the only span with complete data in 2026). Left: average trips per observed day, pooled over January–May, rising from 14,788 in 2024 to 16,120 in 2025 (+9.0%) and 16,871 in 2026 (+4.7%). Right: the same comparison per month, with 2026 above 2025 above 2024 in every one of the five months.

8.13 Reinforcement Learning: Environment Assumptions

Table 17: Environment assumptions for the charging task

Quantity	Value	Meaning
Battery capacity C	{40, 60, 77} kWh	uniform per episode – compact, mid-size, and large vehicle classes
Initial SOC	$\mathcal{U}$(20%, 60%) of C	varies daily with shift length and route profile
Charging window	8 × 15 min	14:00-16:00
Charging powers	{0, 5, 11, 22} kW	zero / low / medium / high (later refined, see Markov Decision Process)
Energy demand d	$\mathcal{N}(30, 5^2)$ kWh, clipped at C	trip demand depends on the route, not on vehicle size – revealed only at departure
Penalty	100 €/kWh shortage	shortfall penalty

8.14 Reinforcement Learning: Results

Table 18: Effect of action-grid resolution in the stylized environment, 1,000 episodes

Grid	Policy	Success rate	Mean reward
Original (4 actions)	Greedy	98.9%	-2.581
Original (4 actions)	Price-unaware DQN	95.5%	-17.201
Refined (7 actions)	Greedy	98.9%	-2.058
Refined (7 actions)	Price-unaware DQN	98.3%	-1.855

Table 19: Policy comparison under EPEX prices, refined action grid

Policy	Success rate	Mean cost (€/day)	Mean reward
Greedy	98.9%	1.7223	-3.7775
Price-unaware DQN (stylized-trained)	98.3%	2.0424	-3.8872
EPEX DQN	75.2%	0.9931	-88.0377
Oracle (40 episodes)	100%	0.87	–

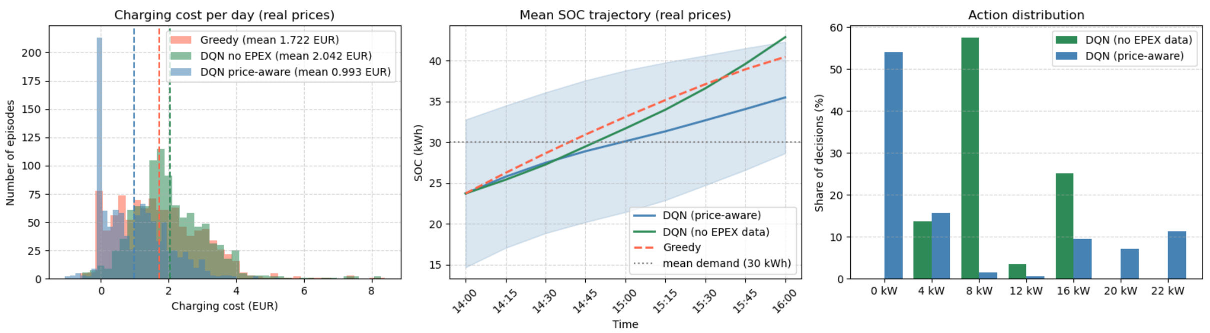

Figure 30: Charging-cost distribution, mean SOC trajectory, and action mix under EPEX prices, refined grid, 1,000 paired episodes. The EPEX DQN's cost mass sits near zero and its SOC trajectory runs lowest of the three policies, consistent with its lower success rate in Table 19; the price-unaware DQN concentrates on 8 kW and 16 kW, while the EPEX DQN uses more of the grid, including 0 kW.

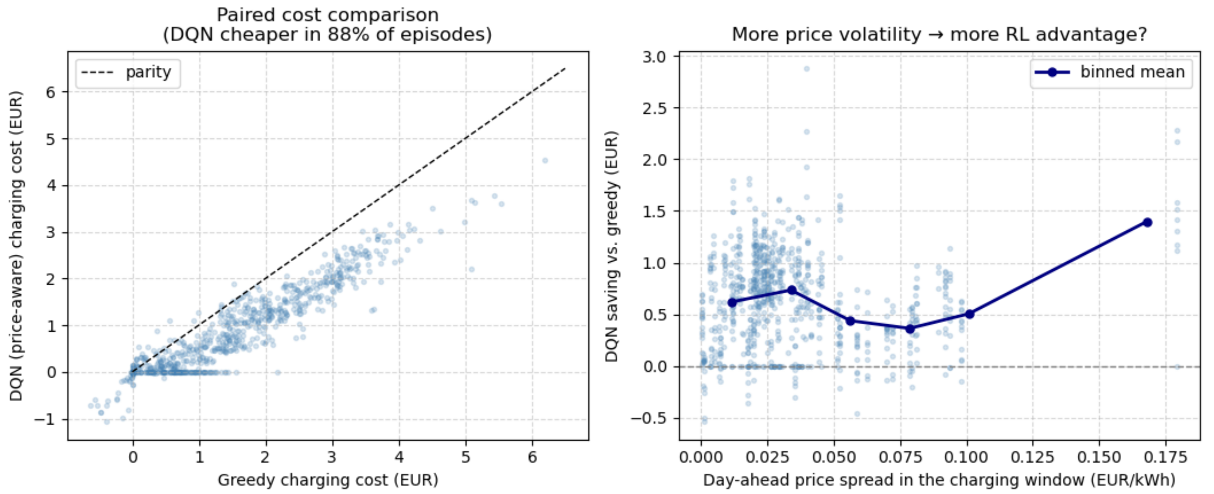

Figure 31: Left: paired charging cost, EPEX DQN vs. greedy, on the 752 episodes both cover – points below the parity line are episodes where the EPEX DQN is cheaper. Right: the EPEX DQN's saving over greedy grows with the day-ahead price spread within the charging window, indicating that the advantage comes from price timing rather than a fixed offset.

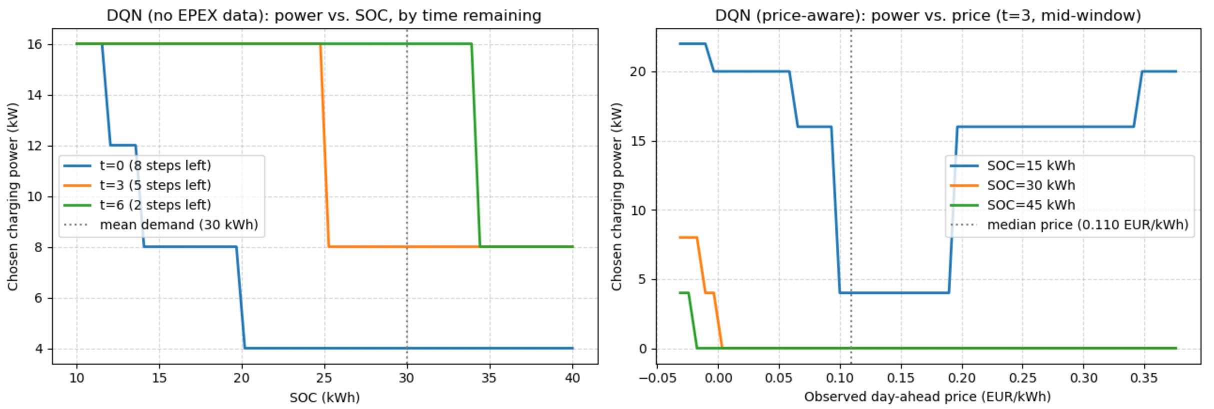

Figure 32: Learned charging policies as a function of state, refined grid. Left: the price-unaware DQN's chosen power falls as SOC rises and as fewer steps remain, independent of price. Right: the EPEX DQN's chosen power falls as price rises above the median (dotted line) and shifts up as SOC drops from 45 to 15 kWh, so urgency dominates price once charge is low.

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